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GROUP SPECIFICATION

Zero-touch network and Service Management (ZSM); Architectural Enhancements for Agent-based Network and Service Management

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Contents

Intellectual Property Rights 6
Foreword 6
Modal verbs terminology 6
1 Scope 7
2 References 7
2.1 Normative references 7
2.2 Informative references 7
3 Definition of terms, symbols and abbreviations 8
3.1 Terms 8
3.2 Symbols 8
3.3 Abbreviations 8
4 Overview of agent technologies for ZSM 9
4.1 Agent and multi-agent concepts in network and service management 9
4.1.1 Overview 9
4.1.2 Requirements 10
4.2 Agent roles in zero-touch management loops 10
4.2.1 Overview 10
4.2.2 Requirements 10
4.3 Design Principles 11
4.3.1 Overview 11
4.3.2 Hierarchical Autonomy 11
4.3.3 Adaptive Learning 11
4.3.4 Intent Driven 11
4.3.5 Protocol Agnosticism 11
4.3.6 Comprehensive Optimal Decision 11
4.3.7 Operation Authorization 11
4.3.8 Explainability and Auditability 12
4.3.9 Requirements 12
5 ZSM framework reference architecture with AI agents 12
5.1 Architectural Overview 12
5.2 Agent onboarding: registration and discovery 13
5.2.1 Overview 13
5.2.2 Agent Onboarding Components 14
5.2.2.1 Agent Registry 14
5.2.2.2 Agent Capability Profile 14
5.2.2.3 Agent Functional Capabilities 14
5.2.2.4 Agent Lifecycle 14
5.2.3 Process: Manage agent registration 15
5.2.3.1 Description 15
5.2.3.2 Procedure flow 16
5.2.3.3 Related management services 18
5.2.4 Process: Manage agent discovery 18
5.2.4.1 Description 18
5.2.4.2 Procedure flow 19
5.2.4.3 Related management services 20
5.2.5 Requirements 20
5.3 Agent communication: messages and tasks 21
5.3.1 Overview 21
5.3.2 Agent Communication Components 21
5.3.2.1 Agent Message 21
5.3.2.2 Task 21
5.3.3 Process: Task delegation and negotiation 22
5.3.3.1 Description 22
5.3.3.2 Procedure flow: Communications patterns 22

5.3.3.3 Procedure flow: Task Negotiation....23
5.3.3.4 Related management services....25
5.3.4 Process: Task status update, escalation and completion....26
5.3.4.1 Description....26
5.3.4.2 Procedure flow....27
5.3.4.3 Related management services....28
5.3.5 Requirements....28
5.4 Semantic interoperability between Agents....29
5.4.1 Overview....29
5.4.2 Semantic Interoperability Components....30
5.4.2.1 Shared vocabulary / ontology....30
5.4.2.2 Use of message metadata to support semantic understanding....30
5.4.2.3 Semantic Repository....31
5.4.3 Process: Semantically-enriched task delegation....32
5.4.3.1 Description....32
5.4.3.2 Procedure flow....33
5.4.3.3 Related management services....35
5.4.4 Requirements....36

6 Management domain support for Agents....36
6.1 Overview....36
6.2 Agents in 3GPP Core domain and RAN domain....37
6.3 Agents in IP and Transport domain....37
6.4 Agents in Fixed access domain....37
6.5 Cross-domain and end-to-end agent scenarios....37

Annex A (normative): Management Services....39

A.1 Additional Services for ZSM architecture....39
A.1.1 Overview....39
A.1.2 Agent management service....39
A.1.3 Agent communication service....40
A.1.4 Semantic interoperability service....40

Annex B (informative): Multi-agent ZSM scenarios....42

B.1 End-to-end zero-touch slice lifecycle using multiple agents....42
B.1.1 Use case description....42
B.1.2 Preconditions....42
B.1.3 Interaction flow....42

B.2 Cross-domain fault correlation and remediation....46
B.2.1 Use case description....46
B.2.2 Preconditions....46
B.2.3 Interaction flow....46

B.3 Closed-loop energy optimization....47
B.3.1 Use case description....47
B.3.2 Preconditions....48
B.3.3 Interaction flow....48
B.3.4 Role of the AN ontology....51

Annex C (informative): Examples of semantic data model....52

C.1 Description....52

C.2 Data model for task....52

C.3 Compatibility with Intent data model....53

C.4 Data model for event....54

C.5 Data model for task negotiation....55

C.6 Example of using data model in information exchange....56

Annex D (informative): Change history 57
History 58

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Foreword

This Group Specification (GS) has been produced by ETSI Industry Specification Group (ISG) Zero-touch network and Service Management (ZSM).

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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1 Scope

The present document specifies extensions and new capabilities to support and integrate agent technologies within the ZSM framework reference architecture to enhance end-to-end zero-touch network and service management.

The present document defines use cases involving agents for network and service management as well as multi-agent communication use cases to derive specific requirements.

The present document specifies requirements for the ZSM architecture to enable agent communication, coordination, and knowledge sharing in support of zero-touch management operations.

The present document is a stage 1 and 2 normative specification based on the ZSM reference architecture and references relevant standards and open-source work where appropriate.

The content of the ETSI GR ZSM 020 [i.1] informative study on the utilization of agents for network and service management served as an input to the present document.

Collaboration with other SDOs (e.g. IRTF NMRG, ITU-T SG13, TM Forum) was pursued where appropriate to ensure alignment and synergy.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents are necessary for the application of the present document.

[1] ETSI GS ZSM 002: "Zero-touch network and Service Management (ZSM); Reference Architecture".

[2] ETSI GS ZSM 009-1: "Zero-touch network and Service Management (ZSM); Closed-Loop Automation; Part 1: Enablers".

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The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] ETSI GR ZSM 020: "Zero-touch network and Service Management (ZSM); Study on the Utilization of Agents in Autonomous Networks".

[i.2] IETF RFC 9315: "Intent-Based Networking - Concepts and Definitions".

[i.3] ETSI TS 128 312: "LTE; 5G; Management and orchestration; Intent driven management services for mobile networks (3GPP TS 28.312 Release 19)".

[i.4] TM Forum IG1253: "Intent in Autonomous Networks v1.3.0".

[i.5] IETF RFC 8639: "Subscription to YANG Notifications".

[i.6] A2A Project: "Agent2Agent (A2A) Protocol Specification", version 1.0, Linux® Foundation.

NOTE: Linux® is the registered trademark of Linus Torvalds in the U.S. and other countries.

[i.7] ETSI GR ZSM 021: "Zero-touch network and Service Management (ZSM); Study of ZSM Framework from Automation to Autonomy".

[i.8] ETSI TS 128 530: "5G; Management and orchestration; Concepts, use cases and requirements (3GPP TS 28.530 Release 19)".

[i.9] 3GPP TR 32.801-01: "Study on 6G Management and Orchestration", Release 20.

[i.10] IETF Internet-Draft draft-yang-nmrg-a2a-nm: "Applicability of A2A to the Network Management", work in progress.

[i.11] IETF Internet-Draft draft-wmz-nmrg-agent-ndt-arch: "Network Digital Twin and Agentic AI based Architecture for AI driven Network Operations", work in progress.

[i.12] ETSI GR F5G 032: "Fifth Generation Fixed Network (F5G); F5G Advanced Use Cases; Release 4".

[i.13] TM Forum IG1453: "Agent-to-Agent Protocol for Telecoms v2.1.0".

[i.14] ETSI GS ZSM 007: "Zero-touch network and Service Management (ZSM); Terminology for concepts in ZSM".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in ETSI GS ZSM 007 [i.14] and the following apply:

autonomous agent: intent-driven, autonomous AI entity designed to perceive its environment, analyse information using the existing knowledge, make comprehensive optimal decisions and execute actions to achieve predefined goals described in intents with minimal to no human intervention, while continuously learning and adapting its behaviours based on the changing conditions in the environment

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in ETSI GS ZSM 007 [i.14] and the following apply:

A2A	Agent to Agent
AN	Autonomous Networks
BER	Bit Error Rate
CLA	Closed-Loop Automation
CL	Closed-Loop
FME	Field Maintenance Engineer
FTTR	Fiber To The Room

LLM	Large Language Model
MAS	Multi-Agent System
NMRG	IRTF Network Management Research Group
NPE	Network-side PE
OAM	Operations And Management
OTN	Optical Transport Network
PON	Passive Optical Network
RAN	Radio Access Network
RM-CL	Ready-Made Closed Loops
SLA	Service Level Agreement
SPE	Service provider-end Provider Edge
UPE	User-end Provider Edge

4 Overview of agent technologies for ZSM

4.1 Agent and multi-agent concepts in network and service management

4.1.1 Overview

As defined in clause 5.1 of ETSI GR ZSM 020 [i.1], the defining characteristic of an autonomous agent is intent-driven, autonomous operation with minimal human intervention. This implies that agents realize one or more closed control loops with the combination and chaining of management services (monitoring, analytics, decision-making and execution).

ETSI GS ZSM 009-1 [2] defined closed loops, including how to enable Closed-Loop Automation (CLA) within the ZSM framework. When one or more intents within a management domain affect the same set of managed entities, multiple closed loops may be instantiated within that management domain to fulfil the goals derived from those intents. However, these closed loops, while performing operations, may encounter goal conflicts, action conflicts, or process dependencies, thereby necessitating coordination among them. To solve these conflicts, that document also introduced several coordination approaches, including assigning different priorities to each closed loop, configuring selection policies for a pre-execution coordination service, etc., so as to automatically select the most appropriate combination of action plans for these closed loops.

Adaptability is a key characteristic of an agent, allowing the CL or multiple CLs to operate adaptively. It implies that, when environmental conditions change, the agent is capable not only of adjusting its plans and actions to fulfil the goal of a single CL, but also of performing comprehensive decision-making to automatically coordinate the action plans of multiple CLs to address conflicts among them.

ETSI GR ZSM 020 [i.1] also defined that "*A Multi-Agent System (MAS) is a system composed of multiple interacting, autonomous agents, each capable of perceiving its environment, making decisions, and taking actions to achieve individual or collective goals. These agents can work independently or collaboratively, often engaging in coordination, negotiation, or competition to solve complex problems or accomplish tasks.*"

In the context of network and service management, the agent-based ZSM reference architecture is a Multi-Agent System (MAS) architecture that employs multiple autonomous agents, not as a simple integration or an "add-on" feature. MAS could encompass the mechanisms of communication and information sharing among collaborative agents, enabling them to achieve individual or collective objectives within a single domain or across multiple management domains or services.

4.1.2 Requirements

The following requirements need to be fulfilled by the ZSM framework to enable agent-based network and service management:

Req-4.1.2-01: The ZSM framework shall support autonomous agents as management functions that produce and/or consume ZSM management services.

Req-4.1.2-02: The ZSM framework shall support autonomous agents that realize one or more closed loops through the consumption and production of ZSM management services.

Req-4.1.2-03: The ZSM framework shall support autonomous agents within management domains and the E2E service management domain.

Req-4.1.2-04: The ZSM framework shall support interaction between autonomous agents within and across management domains.

Req-4.1.2-05: The ZSM framework shall support coordination between closed loops realized by an autonomous agent when their goals, actions or execution dependencies may conflict.

Req-4.1.2-06: The ZSM framework shall support interaction and collaboration between autonomous agents to achieve individual or collective management goals.

Req-4.1.2-07: The ZSM framework shall support coordination, negotiation and information exchange between autonomous agents participating in a multi-agent system.

4.2 Agent roles in zero-touch management loops

4.2.1 Overview

ETSI GR ZSM 021 [i.7] defined both proactive and reactive behaviour patterns for agents. An agent can automatically apply its behaviour pattern based on the needs of a specific scenario. When multiple agents are deployed in ZSM hierarchical architecture as E2E service management domain agents and management domain agents, they collaborate to accomplish complex tasks. Typical collaboration patterns can be listed as follows:

- E2E service management domain agent performs proactive behaviour, management domain agent performs reactive behaviour. For example, the E2E service management domain agent proactively detects service degradation and generates a service optimization solution to prevent customer complaints. The management domain agent performs reactive behaviour by implementing the optimization intent.
- E2E service management domain agent performs reactive behaviour, management domain agent performs proactive behaviour. For example, the management domain agent detects network faults or potential risks, and executes self-remediation actions integrating network quality requirements. The E2E service management domain agent reactively correlates multiple inputs after receiving the complaint order from the customer and reports from the management domain agent, and summarizes the resolution report to the customer.
- E2E service management domain agent and management domain agent perform proactive behaviour. Each agent proactively monitors, analyses, balances actionable goals, and coordinates with each other. Negotiation or competition may occur between the agents in this pattern.

4.2.2 Requirements

The following requirements need to be fulfilled by the ZSM framework to enable autonomous-agent roles in zero-touch management loops:

Req-4.2.2-01: The ZSM framework shall support autonomous agents performing proactive and reactive management behaviours.

Req-4.2.2-02: The ZSM framework shall support collaboration between agents operating in different management domains, including the E2E service management domain.

Req-4.2.2-03: The ZSM framework shall support an agent initiating management actions based on proactively detected network or service conditions.

Req-4.2.2-04: The ZSM framework shall support an agent reacting to goals, intents, events or information received from another agent.

Req-4.2.2-05: The ZSM framework shall support coordination and negotiation between agents when their management actions or actionable goals are interdependent.

4.3 Design Principles

4.3.1 Overview

The adoption of agent technologies is fundamental to the ZSM architecture for achieving fully automated network and service management. In this context, the design of agents to implement key concepts like the closed-loop paradigm - conforming to the ZSM framework and reference architecture defined in ETSI GS ZSM 002 [1] - is governed by a set of core principles that ensure autonomy, resilience, scalability, transparency, and trustworthiness. These principles are described in the following clauses.

4.3.2 Hierarchical Autonomy

The architecture defines a multi-agent system that distributes autonomy across management domains (including E2E service management domain and management domains) through the deployment of autonomous agents, where each agent can perform closed-loop operations within its management domain.

NOTE: In typical deployments, an autonomous agent is responsible for executing Ready-Made Closed Loops (RM-CL) defined in ETSI GS ZSM 009-1 [2], and collaborates with other agents to handle cross-domain intents.

4.3.3 Adaptive Learning

The architecture can support near real-time adaptability across dynamically changing environments, ensuring optimal responses to unpredictable and multi-domain system variations.

4.3.4 Intent Driven

In the ZSM framework, intent-driven interactions between the E2E service management domain agent and the management domain agent enable each autonomous agent to autonomously fulfil its own operational and management intents.

4.3.5 Protocol Agnosticism

Payloads and interaction models that represent agent communication content for telecom operation tasks may be carried using A2A [i.6] or other agent interaction protocols. This design decouples the core logic of autonomous agents' collaboration from the underlying communication technologies.

4.3.6 Comprehensive Optimal Decision

Autonomous agents in the ZSM framework can apply comprehensive optimal decision-making capabilities to achieve a globally satisfactory or near-optimal execution strategy under defined constraints, addressing multi-objective and cross-scenario orchestration complexity within their respective management domains.

4.3.7 Operation Authorization

Agents are granted permissions required for their intended goals. Access control mechanisms for agent interactions with system resources are needed, ensuring that agents execute only authorized operations. This prevents privilege violations and malicious misuse, guaranteeing that agent behaviour remains secure and trustworthy.

4.3.8 Explainability and Auditability

The ZSM framework should provide mechanisms for explainability and auditability to ensure trust, accountability, and regulatory compliance. This includes logging relevant processes and information (e.g. intents, actions and human decisions) for traceability and audit purposes, thereby supporting compliance and incident investigation.

4.3.9 Requirements

The following requirements need to be fulfilled by the ZSM framework to support the design principles for agent-based network and service management:

Req-4.3.9-01: The ZSM framework shall support autonomous agents operating within different management domains, where each agent is capable of performing closed-loop management within its assigned management scope.

Req-4.3.9-02: The ZSM framework shall support autonomous agents adapting their behaviour and management actions in response to changes in their operational environment.

Req-4.3.9-03: The ZSM framework shall support intent-driven interaction between autonomous agents.

Req-4.3.9-04: The ZSM framework shall support agent interaction independently of a specific agent interaction protocol.

Req-4.3.9-05: The ZSM framework shall support autonomous agents coordinating multiple objectives and management actions under defined constraints.

Req-4.3.9-06: The ZSM framework shall support authorization mechanisms that restrict autonomous agents to operations for which they have been granted the required permissions.

Req-4.3.9-07: The ZSM framework shall support traceability and auditability of agent interactions and management actions.

Req-4.3.9-08: The ZSM framework shall support mechanisms that enable explanations of agent decisions and actions to authorized consumers.

5 ZSM framework reference architecture with AI agents

5.1 Architectural Overview

The integration of autonomous agents into the ZSM framework facilitates distributed intelligence and end-to-end service automation across complex network environments. As defined in ETSI GR ZSM 020 [i.1], an agent is an intent-driven, autonomous AI entity designed to perceive its environment, analyse information using existing knowledge, and execute actions to achieve predefined goals with minimal human intervention. These agents are distributed across management domains such as RAN, Core, and IP and the E2E service management domain, where they collaborate within a Multi-Agent System (MAS).

Each agent entity architecturally realizes one or multiple Closed-Loop Automation (CLA) cycles, which integrate monitoring, analysis, decision-making, and execution capabilities to maintain network state based on received intents as described in ETSI GS ZSM 009-1 [2]. Agents are management functions within the ZSM framework, meaning they produce and consume management services to interact with other agents or authorized ZSM consumers.

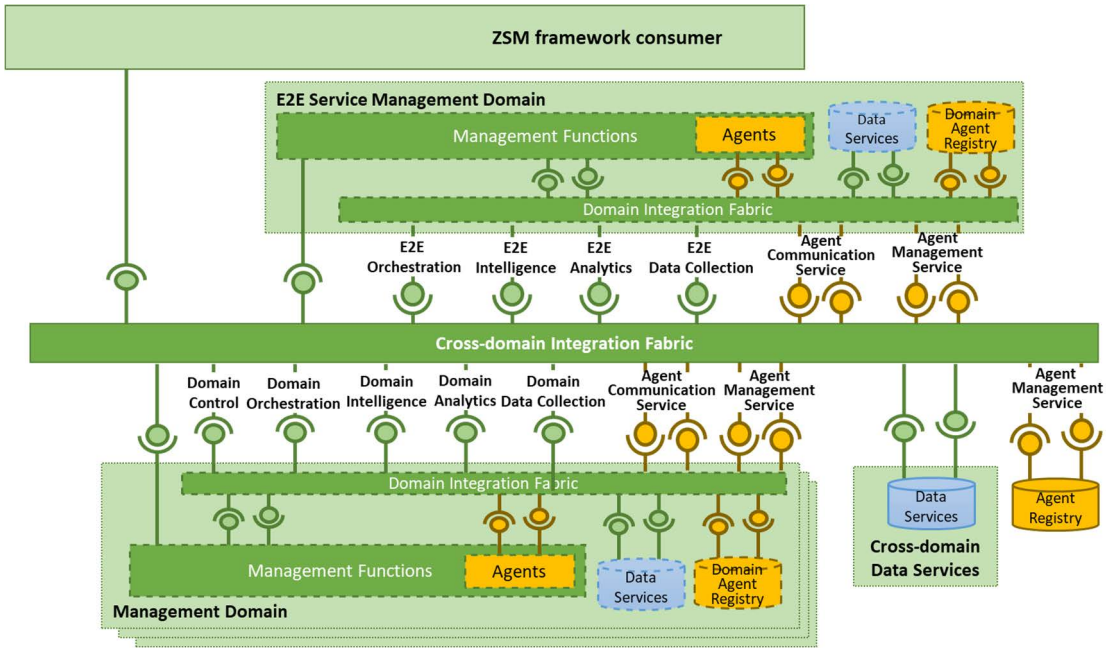


Figure 5.1-1: Agents in the ZSM Framework

Figure 5.1-1 shows that the ZSM integration fabric serves as the primary infrastructure for cross-domain agent interaction, supporting standardized communication through management services defined in the following clauses. A central element of this architecture is the agent registry, a management function that maintains the authoritative catalogue of agent capability profiles. This registry enables the registration, discovery, and lifecycle management of agents, allowing the system to dynamically identify and select the "best" agent for a specific task based on its functional capabilities.

5.2 Agent onboarding: registration and discovery

5.2.1 Overview

Agents need to be registered so they can act as trustworthy management functions within the ZSM framework. Registration provides each agent with a unique identity, links it to a management domain and associates it with security and governance policies. It also exposes the agent's functional capabilities, supported intents, interface endpoints and operational characteristics in a structured form, allowing operators and other components to manage lifecycle aspects such as onboarding, versioning, deprecation and auditing with the same rigor as other ZSM management services.

Agent registration complements the management service registration mechanisms defined in ETSI GS ZSM 002 [1]. As an agent is a management function, the ZSM management services produced by an agent and their associated service capabilities are registered and discovered using the management service registration and discovery mechanisms defined in ETSI GS ZSM 002 [1]. The agent management service defined in the present document additionally manages agent-specific information, including the agent capability profile, functional capabilities, lifecycle information and agent interaction endpoints.

Discovery is needed so that, at runtime, consumers (other agents, orchestrators, or higher-level services) can locate suitable agents in a dynamic, multi-domain environment. By querying the agent registry, as described in ETSI GR ZSM 020 [i.1], they can select the "best" agent for a task based on functional capabilities, policy constraints, performance and trust level, supporting scalable, resilient closed loops. Together, registration (to populate a trusted registry) and discovery (to query and select from it) are essential to enable intent-driven, zero-touch automation where agent-based management functions can be dynamically discovered and orchestrated.

Agent discovery also complements the management services discovery mechanisms defined in ETSI GS ZSM 002 [1]. While management service discovery identifies available ZSM management services and their supported service capabilities, agent discovery identifies agents based on agent-specific characteristics, including their functional capabilities, semantic descriptions, operational characteristics and supported interaction mechanisms.

5.2.2 Agent Onboarding Components

5.2.2.1 Agent Registry

An agent registry is a management function that maintains a catalogue of all registered agents, their identities and their exposed capabilities and makes this information available for discovery and governance within the management framework. It acts as the authoritative source of truth for agent metadata (e.g. identifiers, domains, skills, endpoints, policies, lifecycle state), ensuring that agents are uniquely identifiable, trusted and consistently described.

The agent registry supports core operations such as registering and updating agent descriptions, querying and discovering suitable agents based on functional capabilities and enforcing access control and lifecycle management decisions over the agent population. It therefore underpins both automated agent selection (by other agents or orchestrators) and operational governance (versioning, deprecation, auditing) of the agents in the ZSM framework.

5.2.2.2 Agent Capability Profile

The agent capability profile, also referred to in the industry as an Agent Card, is a structured description of an agent that enables other entities in the ZSM framework to understand what the agent is, what it can do and how to interact with it. It is the formal, machine-readable contract that defines the identity, location and functional capabilities of an agent and it is the primary object registered in the agent registry to enable discovery and interaction within a Multi-Agent System (MAS).

An agent capability profile shall include at least the following components:

- Identity and Management Domain: The unique name of the agent and the specific domain (management domain or E2E service management domain) where it operates.
- Means of Communication: Information identifying how the agent may be reached for agent-to-agent interaction, including applicable service end-points and supported agent interaction protocols.
- Functional Capabilities: A detailed list of tasks or services the agent can perform, such as "Fault Diagnosis", "Quality Optimization", or "Network Slicing".
- Security and trust attributes and authentication requirements for discovery purposes. These security attributes fall outside of the scope of the present document.

5.2.2.3 Agent Functional Capabilities

Agent functional capabilities, also referred to in the industry as agent skills, represent well-defined capabilities of the agent.

These functional capabilities describe the registered behaviour of an agent and are listed on the agent's capability profile so they can be discovered and selected by other agents or clients.

An agent functional capability describes an ability of an agent to perform a particular type of task. It is distinct from the service capability defined in ETSI GS ZSM 002 [1], which represents a specific consumable capability of a ZSM management service. An agent functional capability may be realized through the consumption or production of one or more ZSM service capabilities.

5.2.2.4 Agent Lifecycle

As described in ETSI GR ZSM 020 [i.1], the lifecycle of agents can be structured around two key phases: agent instantiation and agent activation. Agent instantiation is the process of creating the agent entity and preparing it for use, including provisioning its runtime, onboarding it into the management domain and registering its agent capability (identity, endpoints and functional capabilities) in the agent registry. Agent activation is the subsequent process of enabling the instantiated agent to operate so that it can participate in closed loops and pursue its management goals, for example by starting its policies, subscribing to relevant events and becoming available for discovery, task delegation and negotiation.

Beyond instantiation and activation, the agent lifecycle also encompasses operation, evolution, and decommissioning. During operation, the agent executes policies, handles events, and is monitored for performance and health. Evolution allows dynamic updates to configuration, policies, or models in response to environmental changes or self-assessment, potentially triggering partial re-instantiation or re-activation. Decommissioning ensures graceful termination, including de-registration, unsubscription, resource release, and archival of historical data. As noted in ETSI GR ZSM 020 [i.1], these phases are not strictly linear but may form iterative loops, enabling continuous autonomous governance in zero-touch service management.

5.2.3 Process: Manage agent registration

5.2.3.1 Description

The "Manage agent registration" process is the primary onboarding mechanism by which an autonomous agent becomes functional within the ZSM framework. The core of this process is the submission and storage of the agent capability profile in the agent registry. Once the agent capability profile is stored, the agent is available for discovery and can participate in Multi-Agent System (MAS) tasks like delegation and negotiation.

NOTE: As these capabilities are not modelled in ETSI GS ZSM 002 [1], Annex A provides the definitions of these capabilities as an update of the related management services defined in ETSI GS ZSM 002 [1].

5.2.3.2 Procedure flow

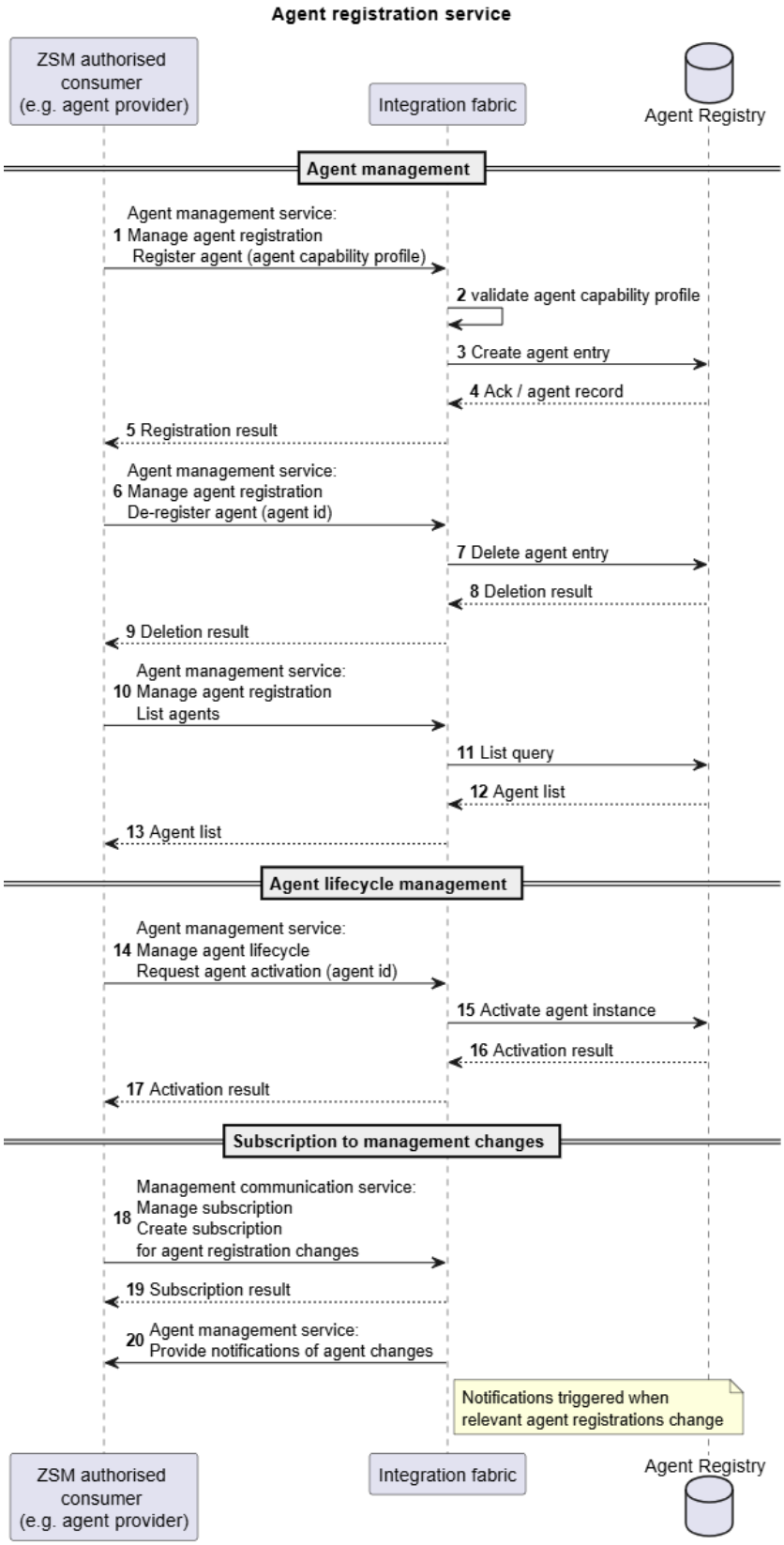


Figure 5.2.3.2-1: Agent registration

PRECONDITIONS:

- None.

The procedure, as illustrated in figure 5.2.3.2-1, consists of the following steps:

Registration:

1) The ZSM authorized agent provider requests the registration, update, or deletion of an agent capability profile by consuming the "Manage agent registration" capability of the "Agent management service" provided by the integration fabric.

2) The integration fabric validates the agent capability profile.

3) The integration fabric creates the corresponding agent entry in the Agent Registry.

4) The agent registry returns an acknowledgement with the stored agent record.

5) The provider receives the registration result.

De-registration:

6) To de-register an agent, the provider invokes the same service with an agent identifier.

7) The integration fabric deletes the agent entry in the registry.

8) The agent registry returns a deletion result.

9) The provider receives the de-registration result.

List:

10) The provider can also list agents by sending the request to the same service.

11) The integration fabric queries the agent registry.

12) The agent registry returns an agent list.

13) The provider receives the agent list.

Agent lifecycle management:

14) The ZSM authorized agent consumer requests activation of an instantiated agent by consuming the "Manage agent lifecycle" capability of the "Agent management service" provided by the integration fabric, using the corresponding agent identifier.

15) The integration fabric triggers activation of the selected agent instance and updates the corresponding lifecycle state in the agent registry.

16) The agent registry returns the status change result.

17) The provider receives the activation result, confirming whether the agent has been successfully activated and is ready for operation.

Subscription to registration changes:

18) The ZSM authorized agent consumer invokes "Manage subscriptions" on the "Management communication service" provided by the integration fabric, requesting notifications about changes in agent registrations or lifecycle changes.

NOTE: The agent consumer may request notifications for agent registration changes from a specific subset of agents by providing their agent identifiers.

19) The integration fabric processes this and returns a subscription result, indicating success or failure of the subscription management operation.

20) Later, when relevant agent registrations are created, updated, or deleted, or the status of agents is modified, the integration fabric sends "notifications about agent changes" to the agent consumer.

POSTCONDITIONS:

- Depending on the requested operation, the agent capability profile is registered, updated, deleted or retrieved.
- Any requested subscription to agent changes is provided.

ERROR CONDITIONS:

- In case the validation of the agent capability profile fails in step 2, the procedure terminates with an error.

5.2.3.3 Related management services

The management services groups and the actual management services (see ETSI GS ZSM 002 [1] and Annex A of the present document) involved in the procedure are summarized below.

The following management services are produced by the ZSM Integration Fabric in this procedure:

- "Manage agent registration", "Manage agent lifecycle" and "Provide notifications of agent changes" capabilities of the "Agent management service" from Annex A.
- "Manage subscriptions" capability of the "Management communication service" from ETSI GS ZSM 002 [1].

5.2.4 Process: Manage agent discovery

5.2.4.1 Description

The "Manage agent discovery" service provided by the integration fabric enables ZSM authorized consumers to identify registered agents through semantic queries on functional capabilities (e.g. "find agents able to perform anomaly detection in transport networks"), using structured metadata, ontological models, and semantic matching to interpret skills, intents, dependencies, and inferred functional capabilities beyond simple keyword searches.

NOTE: As these capabilities are not modelled in ETSI GS ZSM 002 [1], Annex A provides the definitions of these capabilities as an update of the related management services defined in ETSI GS ZSM 002 [1].

5.2.4.2 Procedure flow

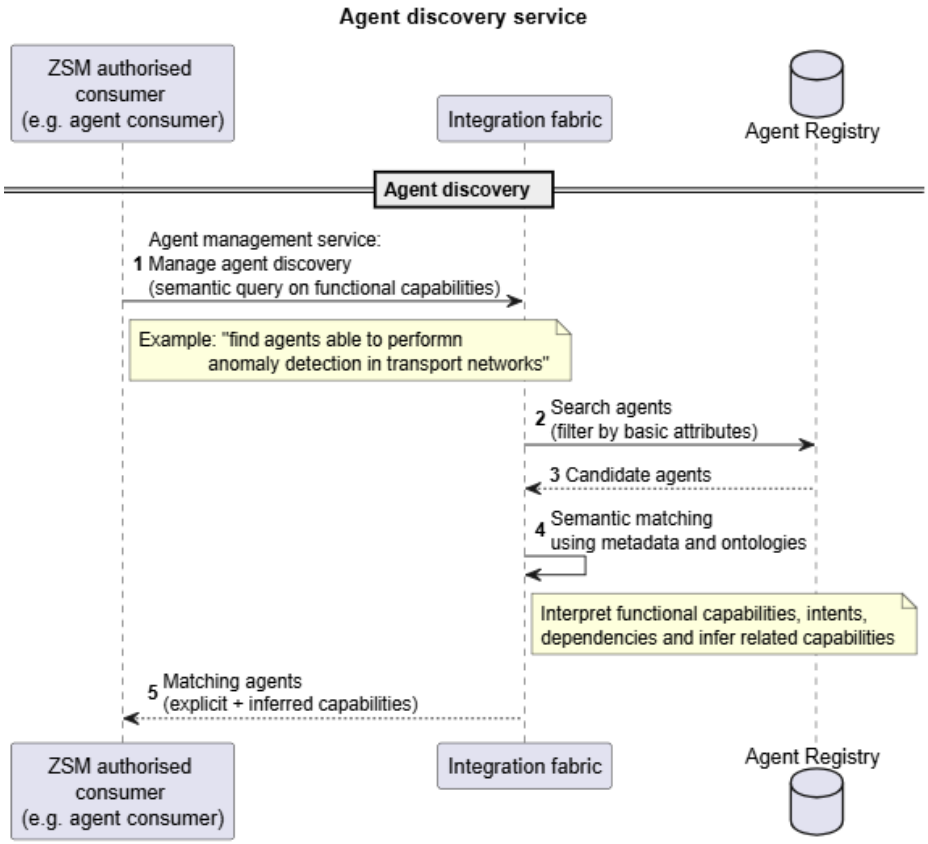


Figure 5.2.4.2-1: Agent discovery

PRECONDITIONS:

- None.

The procedure, as illustrated in figure 5.2.4.2-1, consists of the following steps:

Agent Discovery:

1) The ZSM authorized agent consumer sends a semantic query on agent functional capabilities (e.g. "find agents able to perform anomaly detection in transport networks") to the "Manage agent discovery" capability of the "Agent management service" provided by the integration fabric.

2) The integration fabric queries the agent registry using basic attribute filters to retrieve candidate agents.

3) The agent registry returns candidate agents.

4) The integration fabric performs semantic matching using agent capability profile metadata, ontologies, intents, dependencies, and inferred functional capabilities.

5) The integration fabric returns the list of matching agents (with explicit and inferred functional capabilities) to the agent consumer.

POSTCONDITIONS:

- The matching agents are successfully retrieved from the registry.

ERROR CONDITIONS:

- None.

5.2.4.3 Related management services

The management services groups and the actual management services (see ETSI GS ZSM 002 [1] and Annex A of the present document) involved in the procedure are summarized below.

The following management service is produced by the ZSM Integration Fabric in this procedure:

- "Manage agent discovery" capability of the "Agent management service" from Annex A.

5.2.5 Requirements

The following requirements need to be fulfilled by the ZSM framework to enable autonomous-agent onboarding, registration, lifecycle management and discovery:

Req-5.2.5-01: The ZSM framework shall support the registration, update, retrieval, listing and de-registration of autonomous agents.

Req-5.2.5-02: The ZSM framework shall support a machine-readable agent capability profile for each registered autonomous agent.

Req-5.2.5-03: The agent capability profile shall provide information identifying the agent and the management domain in which it operates.

Req-5.2.5-04: The agent capability profile shall provide information identifying the agent's functional capabilities.

Req-5.2.5-05: The agent capability profile shall provide information identifying the means by which authorized consumers can interact with the agent, including applicable service end-points and supported agent interaction protocols.

Req-5.2.5-06: The ZSM framework shall support the association of security and trust-related information with an agent capability profile for agent discovery and interaction authorization.

Req-5.2.5-07: The ZSM framework shall support lifecycle management of autonomous agents.

Req-5.2.5-08: The ZSM framework shall support the discovery of registered autonomous agents based on their agent capability profiles.

Req-5.2.5-09: The ZSM framework shall support discovery of registered autonomous agents based on their functional capabilities.

Req-5.2.5-10: The ZSM framework shall support filtering of candidate autonomous agents based on agent capability profile attributes.

Req-5.2.5-11: The ZSM framework should support semantic matching of agent functional capabilities during agent discovery.

Req-5.2.5-12: The ZSM framework should support discovery results based on both explicitly declared and semantically inferred agent functional capabilities.

5.3 Agent communication: messages and tasks

5.3.1 Overview

Agent communication is the fundamental mechanism that enables distributed intelligence within a Multi-Agent System (MAS). It serves as the foundation for autonomous agents to interact, collaborate, and coordinate their actions to solve complex problems or achieve collective goals. Unlike simple data transmission, communication in this framework allows agents to establish shared situational awareness, align operational objectives, and maintain coherence in decision-making across different management domains. Therefore, in an autonomous agent-based ZSM framework, the design of interface interactions for agent communication should consider several aspects:

- **Declarative:** Interfaces are required to be declarative, allowing a consumer to specify the goal (what) rather than the explicit sequence of steps (how). This leaves room for the provider agent to evolve itself to fulfil the goal continuously.
- **Compatibility:** Interfaces are required to remain backward compatible, and interoperable when carrying, as payload, intent-related data models originating from different Standards Development Organizations (SDOs).
- **Efficiency:** Interfaces for agent-to-agent communication should be designed for efficiency, for example, by employing semi-structured data formats that enable deterministic expression of message payloads, instead of depending on natural language intended solely for human interpretation.
- **Simplicity**: Interfaces should be abstract and simple, conveying adequate information and retaining the agent's internal implementation details.

Effective communication is critical for the ZSM framework to support intent-driven automation, where agents negotiate task fulfilment, share learned knowledge, and adapt to changing environment conditions.

NOTE: Agent communication may use either direct or indirect service interaction patterns as defined in ETSI GS ZSM 002 [1]. In direct interaction, an agent communicates with another agent through a discovered service end-point. In indirect interaction, communication is mediated by the integration fabric.

5.3.2 Agent Communication Components

5.3.2.1 Agent Message

An agent message is the standardized unit of information exchange used for interaction between autonomous entities. Messages are the vehicle for sharing information to achieve common goals, delegate tasks, resolve conflicts and coordinate actions. Depending on the selected communication model (such as Peer-to-Peer or Publish/Subscribe), messages may be sent to specific peers or disseminated across shared channels to multiple subscribers.

5.3.2.2 Task

In the ZSM framework, a task represents a structured unit of work that one autonomous agent delegates to another. It is a persistent, trackable, and stateful context designed for asynchronous, long-running collaborations, ensuring clarity, precise task definition, and accountability between agents.

Semantic understanding of the task meaning between agents is crucial to ensure that both the delegating and executing agents share a common interpretation of objectives, constraints, and expected outcomes. As described in ETSI GR ZSM 020 [i.1], agent messages can be enriched with semantic attributes such as key-value pairs defining specific terms or with references to external resources, including files or ontologies, where these definitions are maintained. By grounding task exchanges in shared semantics agents can more effectively reason about intent, context and task interdependencies. This enables interoperability across management domains within the ZSM framework.

Key aspects of a task include:

- Task negotiation: Unlike static rules, tasks are often distributed through a negotiation process where agents propose, accept or decline based on their functional capabilities, current workload, and performance history.
- Delegation and escalation: Agents can break down complex workflows into discrete tasks and delegate them to specialized entities (e.g. a "Fault Diagnosis" task assigned to a specific domain agent).

- Agreed parameters: A task encompasses formally accepted parameters, including requirements for time, quality and resource use.

5.3.3 Process: Task delegation and negotiation

5.3.3.1 Description

The "Provide agent message" process enables reliable, standardized communication between autonomous agents within the ZSM framework using agent messages and tasks as the fundamental unit of interaction. This process supports both peer-to-peer and publish-subscribe communication patterns, allowing agents to exchange information, intents, knowledge, and coordination signals across management domains to achieve collective management goals.

NOTE: As these capabilities are not modelled in ETSI GS ZSM 002 [1], Annex A provides the definitions of these capabilities as an update of the related management services defined in ETSI GS ZSM 002 [1].

5.3.3.2 Procedure flow: Communications patterns

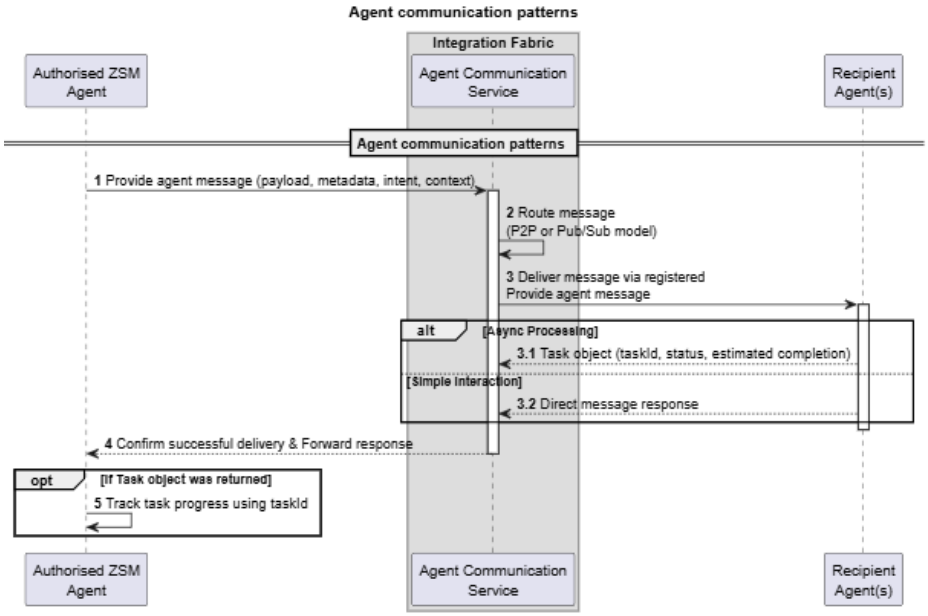


Figure 5.3.3.2-1: Provide agent message: communications patterns

PRECONDITIONS:

- Sender and recipient agents are registered in the agent registry.
- Sender has discovered recipient agent.
- Recipient endpoint from agent capability profile is accessible.

The procedure, as illustrated in figure 5.3.3.2-1, consists of the following steps:

Message transmission patterns:

1) The authorized ZSM agent constructs an agent message containing payload, metadata, intent, and context, then consumes the "Provide agent message" capability of the "Agent Communication Service" targeting one or more recipient agents via their registered endpoints.

2) Depending on the applicable interaction pattern, the message is delivered either directly to the recipient agent through its discovered service end-point or indirectly through the integration fabric. For publish-subscribe communication, the integration fabric mediates delivery using the applicable communication channel.

3) Recipient agent(s) receive the message through their communication endpoint and respond immediately with either:

    1) Task object (taskId, status, estimated completion) for async processing; or

    2) Direct Message response for simple interactions.

4) The sender receives the response (task or message) and if a task was returned, tracks its progress using the task identifier.

5) The integration fabric confirms successful delivery to the sender or notifies of delivery failure.

POSTCONDITIONS:

- Interaction initiated with immediate acknowledgement; processing may continue asynchronously if Task returned.

ERROR CONDITIONS:

- If message delivery fails (e.g. endpoint unreachable, authentication failure, policy violation), the management service returns an error to the sender and may trigger retry mechanisms or escalation procedures.

5.3.3.3 Procedure flow: Task Negotiation

PRECONDITIONS:

- Sender and recipient agents are registered in the agent registry.
- Sender has discovered recipient agent.
- Recipient endpoint from agent capability profile is accessible.

The procedure, as illustrated in figure 5.3.3.3-1, consists of the following steps:

Task delegation and negotiation via initial message:

1) The authorized ZSM agent constructs an initial agent message containing an embedded task definition (goals, constraints, contextId), then consumes the "Provide agent message" capability of the "Agent Communication Service" targeting the recipient agent via its registered service end-point from the Agent Registry.

2) Recipient evaluates task proposal against its capabilities, workload, and context.

3) The recipient agent can now:

    1) Accept: Returns task object (status: "accepted", estimated completion).

    2) Propose counter-offer: Returns task object (status: "proposed", alternative parameters).

4) Sender agent evaluates proposal parameters and decides next action.

5) The sender agent may accept the counter-proposal or provide a further counter-proposal:

    1) Accept counter-proposal: Sends acceptance message (taskId); or

    2) Sender agent sends counter-counter-proposal;

    3) Recipient agent may accept, provide a counter-proposal, or decline the task.

6) Sender may also decline the updated proposal from Recipient.

7) Task execution starts once status reaches "accepted"; recipient performs async processing.

8) Recipient informs sender that processing starts.

9) Sender tracks task progress via status updates (in-progress/completed).

10) Recipient notifies completion with final results or escalates if needed.

POSTCONDITIONS:

- Task negotiation concludes with one of the following outcomes:
  - Recipient accepts the initial or a negotiated Task, sets status to accepted or in-progress, and begins processing.
  - Sender declines or negotiation times out, resulting in rejected or cancelled state.
- Sender holds a reference to the task (taskId) and starts tracking progress if the task is accepted.
- Both agents have common understanding of the task's state and agreed parameters (goals, deadlines, quality metrics, etc.).
- The Agent Communication Service logs the message exchange outcome for auditing and traceability.

ERROR CONDITIONS:

- Message delivery failure: Recipient endpoint unreachable, timeout, or routing error in the Agent Communication Service; sender receives delivery-failure notification or retry recommendation.
- Authentication/authorization failure: Sender not authorized to use the "Provide agent message" capability or recipient rejects message due to invalid credentials.
- Malformed or invalid task data: Missing mandatory fields such as goals or contextId, causing message rejection and error return to sender.
- Negotiation timeout: Recipient or sender fails to respond within agreed SLA window; task automatically transitions to expired or failed-negotiation state.
- Processing failure after acceptance: Recipient acknowledges task but later reports error (e.g. resource unavailability, internal exception), updating task state to failed and notifying the sender.

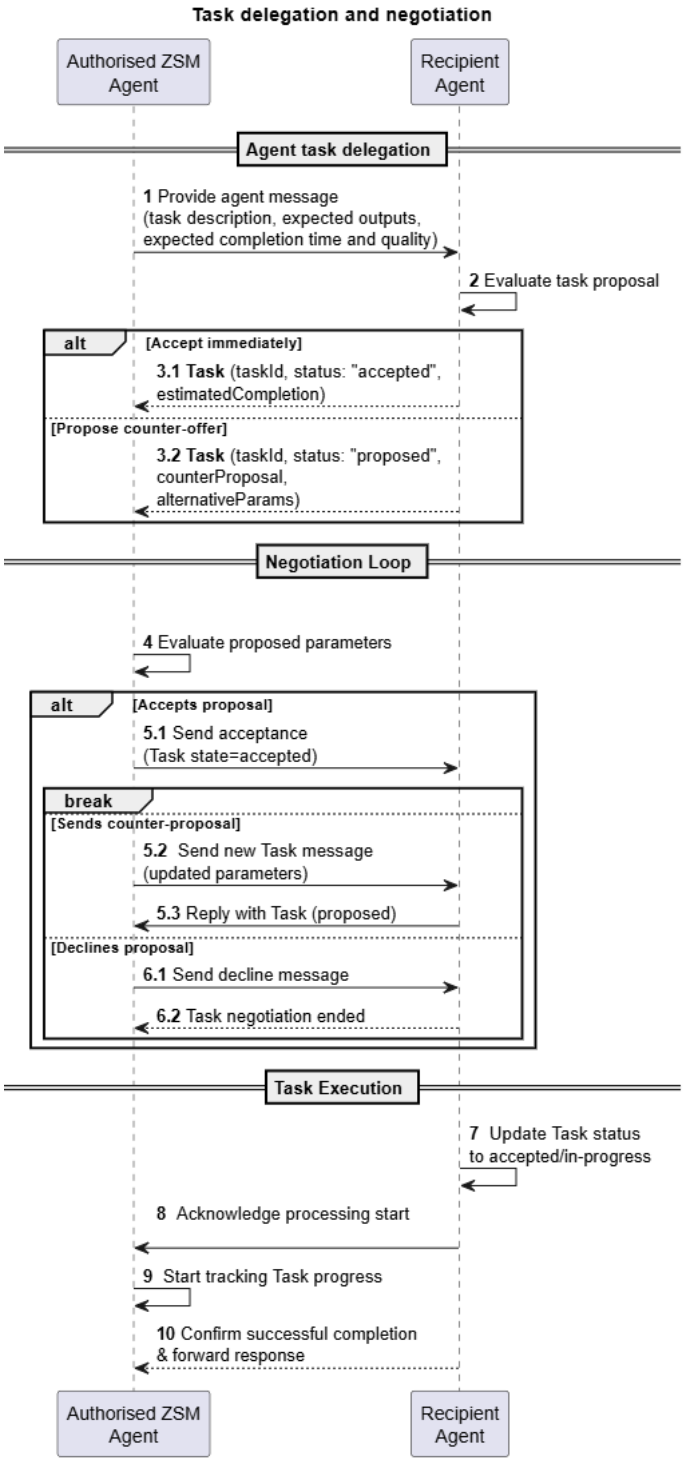


Figure 5.3.3.3-1: Provide agent message: task delegation and negotiation

5.3.3.4 Related management services

The management services groups and the actual management services (see ETSI GS ZSM 002 [1] and Annex A of the present document) involved in the procedure are summarized below.

The following management service is produced by management functions within the management domain and the E2E service management domain in this procedure:

- "Provide agent message" capability of the "Agent communication service" from Annex A.

5.3.4 Process: Task status update, escalation and completion

5.3.4.1 Description

The Manage task process enables full lifecycle management of delegated tasks using the "Manage task" capability and "Provide task status" notifications of the Agent Communication Service. The delegating agent can read task status, while executing agents provide real-time progress updates, trigger escalation for issues beyond their scope, and report completion with outcomes. This ensures continuous visibility and governance of asynchronous task execution.

NOTE: As these capabilities are not modelled in ETSI GS ZSM 002 [1], Annex A provides the definitions of these capabilities as an update of the related management services defined in ETSI GS ZSM 002 [1].

5.3.4.2 Procedure flow

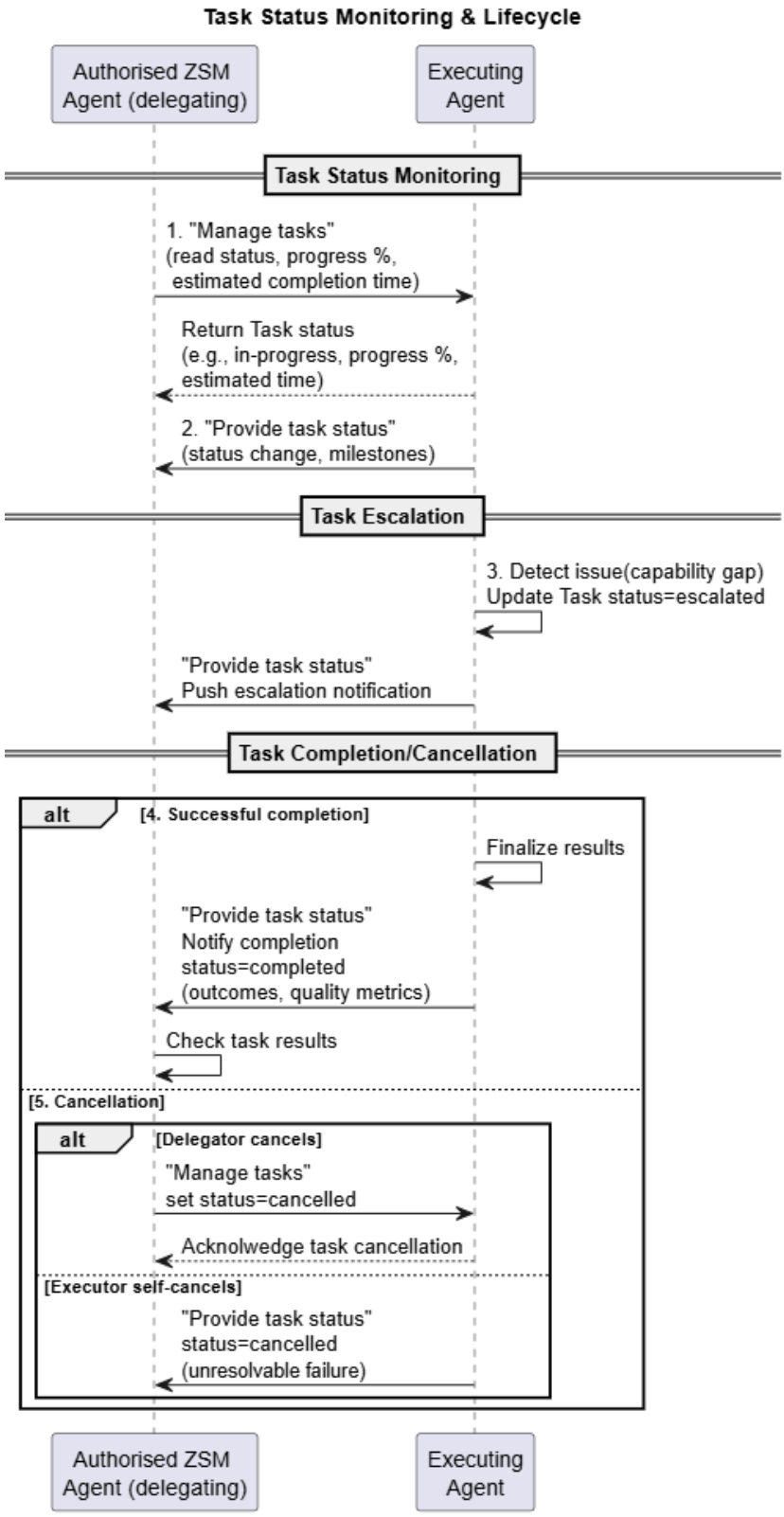


Figure 5.3.4.2-1: Task status update, escalation and completion

PRECONDITIONS:

- Task has been successfully delegated and accepted (status: accepted/in-progress).

- Sender agent has taskId references.
- Both agents are registered and authorized for task communication.

The procedure, as illustrated in figure 5.3.4.2-1, consists of the following steps:

Task Status Monitoring:

1) Pull-based query: The authorized ZSM agent (delegating) consumes "Manage task" capability to read task status (e.g. in-progress, progress percentage, estimated completion).

2) Push-based notification: Executing agent consumes "Provide task status" capability to proactively notify delegating agent of status changes (milestones reached, delays detected).

Task Escalation:

3) Executing agent detects an issue beyond its scope (e.g. capability gap), updates the status of the task to escalated, terminates its processing of the task, and notifies the delegating agent. The delegating agent may subsequently create or delegate a new task to another agent or initiate an alternative resolution process.

Task Completion/Cancellation:

4) Successful completion: Executing agent finalizes results, consumes "Provide task status" with status completed, including outcomes and quality metrics; delegator acknowledges receipt.

5) Cancellation: Delegator consumes "Manage task" to cancel task (status: cancelled), or executing agent self-cancels due to unresolvable failure; both parties receive final notification.

POSTCONDITIONS:

- Task reaches terminal state (completed, cancelled, failed, escalated). In the escalated state, execution of the current task has terminated and further handling is performed through a subsequent task or process.
- All agent actions, logs, and outcomes preserved for audit.
- Delegating agent can trigger follow-on actions or terminate the closed-loop process.

ERROR CONDITIONS:

- TaskNotFoundError: Invalid taskId during status query or update.
- UnauthorizedAccessError: Agent lacks permission to update/escalate specific task.

5.3.4.3 Related management services

The management services groups and the actual management services (see ETSI GS ZSM 002 [1] and Annex A of the present document) involved in the procedure are summarized below.

The following management service is produced by management functions within the management domain and the E2E service management domain in this procedure:

- "Manage task" and "Provide task status" capabilities of the "Agent communication service" from Annex A.

5.3.5 Requirements

The following requirements need to be fulfilled by the ZSM framework to enable communication and task-based collaboration between autonomous agents:

Req-5.3.5-01: The ZSM framework shall support standardized message exchange between autonomous agents.

Req-5.3.5-02: The ZSM framework shall support agent messages carrying goals, intents, context, metadata and task-related information.

Req-5.3.5-03: The ZSM framework shall support agent communication using direct and indirect service interaction patterns.

Req-5.3.5-04: The ZSM framework shall support both peer-to-peer and publish-subscribe communication between autonomous agents.

Req-5.3.5-05: The ZSM framework shall support the delegation of structured tasks between autonomous agents.

Req-5.3.5-06: The ZSM framework shall support persistent and uniquely identifiable tasks for asynchronous agent collaboration.

Req-5.3.5-07: The ZSM framework shall support lifecycle state information for delegated tasks.

Req-5.3.5-08: The ZSM framework shall support an agent accepting or declining a delegated task.

Req-5.3.5-09: The ZSM framework shall support negotiation of task parameters between requesting and executing agents.

Req-5.3.5-10: The ZSM framework shall support asynchronous execution of an accepted task.

Req-5.3.5-11: The ZSM framework shall support an authorized agent querying the status of a delegated task.

Req-5.3.5-12: The ZSM framework shall support notification of task status changes to authorized consumers.

Req-5.3.5-13: The ZSM framework shall support escalation of a task when the executing agent cannot complete the task within its operational scope.

Req-5.3.5-14: The ZSM framework shall support cancellation of delegated tasks.

Req-5.3.5-15: The ZSM framework shall support reporting of task completion together with the corresponding task outcome.

Req-5.3.5-16: The ZSM framework shall support reporting of task failure and associated failure information.

Req-5.3.5-17: The ZSM framework shall support traceability of agent message exchanges and task lifecycle transitions.

5.4 Semantic interoperability between Agents

5.4.1 Overview

Semantic interoperability enables agents belonging to different management domains to exchange goals, intents, context, observations, constraints and outcomes without losing the intended meaning of the information exchanged.

In a multi-agent ZSM framework, agents may expose similar capabilities using different terminology, data models, abstractions or domain-specific vocabularies. Semantic interoperability provides the common interpretation layer required for agents to align these meanings before delegating tasks, negotiating goals, exchanging context, or correlating events. As described in ETSI GR ZSM 020 [i.1], agents may encounter semantic misinterpretations during task collaboration when terms carry different meanings across domains (e.g. "CIR" meaning "Carrying Information Rate" in one domain and "Committed Information Rate" in another, or "RAN" meaning "Radio Access Network" in a RAN-domain agent and "Remote Area Network" in an IP-domain agent).

Semantic interoperability is supported through:

- shared vocabularies, ontologies and semantic mappings
- machine-readable metadata and semantic references attached to agent capabilities, tasks, intents, events and outcomes
- semantic validation of exchanged information
- translation or mapping between domain-specific and cross-domain information models
- multi-round clarification when semantic ambiguity cannot be resolved automatically

The semantic interoperability mechanisms described in this clause define the information and process requirements for exchanging meaning between agents and may be realized over different agent communication protocols, management service interfaces, message buses, eventing mechanisms, or integration fabric capabilities.

5.4.2 Semantic Interoperability Components

5.4.2.1 Shared vocabulary / ontology

A shared vocabulary or ontology is a set of common semantic assets that enables agents in a multi-agent ZSM environment to assign consistent meaning to the terms, concepts, relationships, and structures used in exchanged information. These assets form the foundational layer of semantic interoperability by providing a common frame of reference that agents may use when expressing or interpreting goals, intents, tasks, events, context, constraints, and outcomes.

Shared semantic assets may include the following types:

- **Controlled vocabularies:** Defined sets of terms with agreed meanings covering ZSM domains, such as service management, network management, resource management, fault management, performance management, and energy management. Controlled vocabularies reduce ambiguity by limiting the set of acceptable terms used in agent interactions and by associating each term with a normative definition.

- **Ontologies:** Formal representations of concepts, their attributes, and the relationships between them within a given domain or across domains. Ontologies allow agents to reason about the meaning of exchanged content, to infer relationships between concepts, and to identify equivalences or conflicts between terms used by agents from different management domains. An ontology may represent concepts such as network slices, managed objects, service functions, performance indicators, fault classes, or policy conditions.

- **Taxonomies:** Hierarchical classification schemes that organize agents, capabilities, event types, object types, context categories, or managed entities into structured categories. Taxonomies support consistent filtering, routing, and matching of agent capabilities, goals, and events by enabling agents to determine whether two concepts belong to the same class or category, even when expressed using different terminology.

- **Semantic mappings:** Definitions of equivalences, translations, or correspondences between terms, concepts, or structures used in different domain-specific vocabularies or information models. A semantic mapping may express, for example, that the term used by a RAN-domain agent for a specific managed object corresponds to the term used for the same object by a core-network-domain agent, or that a service-level intent expressed using one vocabulary corresponds to a set of resource-level parameters expressed in another. Mappings may carry a confidence level or quality indicator reflecting the degree of equivalence.

NOTE: Different intent models could serve as the foundation for the shared vocabulary or ontology, such as the IETF [i.2], 3GPP [i.3] or TM Forum [i.4] intent models.

5.4.2.2 Use of message metadata to support semantic understanding

Agents may include semantic context in exchanged messages to help the receiving agent interpret the meaning of natural-language, structured-data or hybrid content. This semantic context may be carried as metadata associated with the main message depending on the communication mechanism used.

NOTE: Metadata may be used with any type of message content. However, it becomes particularly important when messages contain natural language and the intended meaning may depend on additional context and terminology.

The metadata does not replace the actual task, goal, intent, event or result content. Instead, it provides interpretation hints and semantic references that allow the receiver to understand the content consistently. This is especially useful when the exchanged content contains natural language, domain-specific terminology, ambiguous goals, model-specific field names, or references to external standards.

Examples of semantic metadata may include:

- semantic model identifier and version

- vocabulary, ontology or taxonomy references

- source domain and target domain
- task, goal, intent or event type
- managed object type and identifier scheme
- context type, such as topology, time, location, customer, service, slice, fault, performance or energy context
- metric definitions and units
- constraint semantics
- expected output semantics
- mapping references between source and target concepts
- confidence level of an inferred or mapped meaning

Use of metadata for semantic understanding should follow these principles:

- semantic metadata should be optional unless required by policy or by the target agent
- semantic metadata should not conflict with the actual message or task
- semantic metadata should identify the semantic model or vocabulary version when versioning affects interpretation
- semantic metadata should be machine-readable where possible
- semantic metadata should support clarification when the receiver cannot determine the intended meaning
- sensitive semantic context should be protected according to the applicable security and authorization policies

This approach keeps the semantic interoperability process protocol agnostic while allowing management services implementations to reuse the existing metadata mechanism for practical realization.

5.4.2.3 Semantic Repository

A semantic repository is an optional management function that stores and exposes semantic assets used by agents to interpret exchanged information consistently. These semantic assets may include shared vocabularies, ontologies, taxonomies, controlled terms, semantic mappings, model references, metric definitions, unit definitions, policy concepts, and domain-specific information model references.

The semantic repository is optional because semantic interoperability can also be achieved without a centralized repository. For example, agents may carry semantic references directly in their capability profiles, exchange semantic hints in message metadata, rely on locally configured mappings, or use bilateral agreements between domains. A semantic repository is therefore not a mandatory architectural component; rather, it is a deployment option that becomes useful when semantic assets need to be governed, reused, versioned, audited, or shared across multiple agents, management domains or administrative boundaries.

The absence of a semantic repository does not prevent agents from achieving semantic interoperability. In such cases, the required semantic context should be exchanged directly between agents or made available through the agent capability profile, message metadata, preconfigured mappings, or domain-specific management services.

5.4.3 Process: Semantically-enriched task delegation

5.4.3.1 Description

The "Semantically-enriched task delegation" process extends the "Task delegation and negotiation" process defined in clause 5.3.3 by augmenting the "Provide agent message" capability of the "Agent communication service" with ontology references and semantic metadata. The extension enables a requesting agent and a target agent operating in different management domains, potentially using different large language models or domain-specific terminologies, to reach a shared interpretation of the task goal, managed object scope, constraints, context, and expected outputs before and during task execution.

The process supports three complementary mechanisms for achieving semantic alignment:

- **Inline semantic metadata:** The requesting agent includes semantic context directly in the agent message as structured metadata (e.g. vocabulary reference, ontology identifier, managed object type, metric definitions, constraint semantics), as described in clause 5.4.2.2. This mechanism is always available and does not require a semantic repository.
- **Semantic repository resolution:** The requesting agent or the integration fabric resolves semantic references using the "Resolve semantic reference" capability of the "Semantic interoperability service" before or during message routing. The target agent may also invoke this capability independently to interpret received content. This mechanism is applicable when a semantic repository is deployed.
- **Semantic clarification exchange:** When the target agent cannot resolve the meaning of a received goal, intent, constraint or context item from the supplied metadata or from the semantic repository, it initiates a clarification exchange with the requesting agent using the "Provide agent message" capability, requesting the missing semantic information. The requesting agent responds with the required clarification, which may include additional metadata, vocabulary references, or inline term definitions.

NOTE: The process defined in this clause does not modify the structure of the "Provide agent message" capability defined in clause 5.3. Semantic metadata is carried as additional, optional attributes of the agent message object alongside the task definition, goal and context objects already defined in clause 5.3.2.

5.4.3.2 Procedure flow

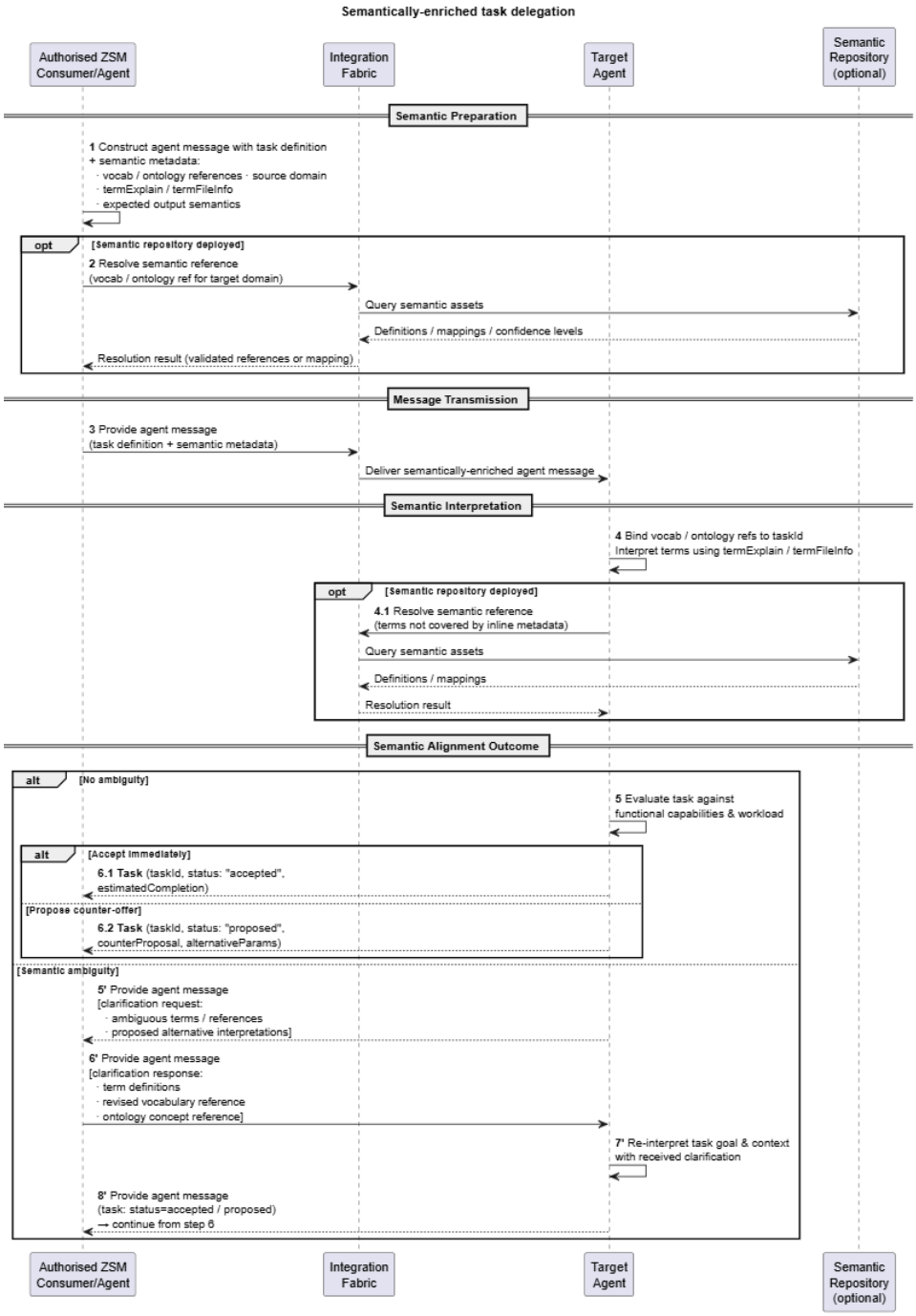


Figure 5.4.3.2-1: Semantically-enriched task delegation

PRECONDITIONS:

- Requesting and target agents are registered in the agent registry.
- Requesting agent has discovered the target agent using the "Manage agent discovery" capability of the "Agent management service".
- Target agent endpoint from the agent capability profile is accessible.
- If a semantic repository is deployed: the requesting agent has access to the "Semantic interoperability service" provided by the integration fabric.

The procedure, as illustrated in figure 5.4.3.2-1, consists of the following steps:

Semantic preparation (requesting agent):

1) The requesting agent constructs an agent message containing a task definition and enriches it with semantic metadata as described in clause 5.4.2.2. The semantic metadata may include:

    1) References to shared vocabulary or ontology identifiers applicable to the task goal and managed object.

    2) The source management domain of the requesting agent.

    3) Key-value term definitions for domain-specific terms used in the task goal (e.g. termExplain: [{"CIR": "Committed Information Rate"}]) or a reference to a term file (termFileInfo).

    4) Expected output semantics and applicable metric definitions.

2) (optional – when a semantic repository is deployed): Before sending the message, the requesting agent or the integration fabric invokes the "Validate semantic reference" capability of the "Semantic interoperability service" to verify the semantic references and, where resolution or mapping is required, invoke "Resolve semantic reference". The integration fabric may perform this resolution transparently as part of message routing.

2') (optional – when a semantic repository is deployed): If the semantic assets referenced by the task may be updated during the task lifecycle, the requesting agent may consume the "Manage subscriptions" capability of the "Management communication service" defined in ETSI GS ZSM 002 [1] to subscribe to changes affecting those semantic assets. When a subscribed semantic asset is updated, deprecated or otherwise changed, the "Provide semantic asset notifications" capability of the "Semantic interoperability service" notifies the subscribed agent.

Message transmission with semantic enrichment:

3) The authorized ZSM requesting agent consumes the "Provide agent message" capability of the "Agent communication service", sending the semantically-enriched agent message to the target agent via its registered endpoint.

Semantic interpretation (target agent):

4) The target agent receives the agent message and attempts to interpret the task by applying the semantic metadata supplied in the message:

    - The target agent binds the vocabulary and ontology references to the task identified by the taskId and interprets domain-specific terms in the goal and context using the supplied term definitions or term file references.

    - If a semantic repository is deployed, the target agent may additionally invoke the "Resolve semantic reference" capability of the "Semantic interoperability service" to obtain definitions or mappings for any terms not covered by the inline metadata.

5) **No ambiguity:** If the target agent can fully interpret the task goal, constraints, and expected outputs, it evaluates the task proposal against its functional capabilities and workload, and responds using the "Provide agent message" capability with one of the following:

    - Task object with status "accepted" and estimated completion; or

    - Task object with status "proposed" carrying a counter-offer with alternative parameters.

Subsequent negotiation, acceptance, and execution steps follow the procedure defined in clause 5.3.3.3.

Semantic clarification exchange (when ambiguity cannot be resolved):

5') **Semantic ambiguity:** If the target agent cannot unambiguously interpret the received goal, intent, constraint or context item, it sends a clarification request to the requesting agent using the "Provide agent message" capability. The clarification request identifies the ambiguous terms or references and may propose alternative interpretations for confirmation.

6) The requesting agent receives the clarification request and responds with a clarification message providing the missing semantic information. The response may include additional inline term definitions, a revised vocabulary reference, or a reference to a specific ontology concept that resolves the ambiguity.

7) The target agent processes the clarification, re-interprets the task goal and context, and proceeds to evaluate and accept or counter-propose the task as in step 5.

POSTCONDITIONS:

- Both agents share a consistent interpretation of the task goal, managed object scope, constraints, expected outputs and applicable metrics.
- The target agent holds a reference to the taskId and begins or confirms processing.
- All semantic metadata exchanged during the interaction is available for audit and traceability.
- Where a subscription to semantic asset changes has been established, the subscribed agent can receive notifications of changes to semantic assets that may affect interpretation of the task.

ERROR CONDITIONS:

- Unresolvable semantic reference: The semantic repository returns an error or unknown result for a requested vocabulary or ontology reference. The integration fabric or the requesting agent may fall back to inline metadata or initiate a semantic clarification exchange.
- Semantic conflict: The target agent identifies an irreconcilable conflict between the semantics declared by the requesting agent and its own domain vocabulary. The target agent declines the task and returns an error indicating a semantic conflict, triggering escalation to a human operator or a semantic mediation function.
- Clarification timeout: If the requesting agent fails to respond to a clarification request within the agreed SLA window, the task transitions to expired or failed-negotiation state.
- Missing mandatory semantic reference: If the task goal or context references a vocabulary or ontology that is not available to the target agent and cannot be resolved, the task is rejected with an error identifying the unresolvable reference.

5.4.3.3 Related management services

The management services groups and the actual management services (see ETSI GS ZSM 002 [1] and Annex A of the present document) involved in the procedure are summarized below.

The following management service is produced by management functions within the management domain and the E2E service management domain in this procedure:

- "Provide agent message" capability of the "Agent communication service" from Annex A. Used for all message exchanges between requesting and target agents, including semantically-enriched task delegation and semantic clarification exchanges.

The following management services are produced by the ZSM Integration Fabric in this procedure:

- "Resolve semantic reference" and "Validate semantic reference" capabilities of the "Semantic interoperability service" from Annex A. Used to resolve vocabulary and ontology references and to retrieve applicable term definitions and mappings from the semantic repository (when deployed).
- "Provide semantic asset notifications" capability of the "Semantic interoperability service" from Annex A. Used by subscribed agents to receive notifications about updates or deprecations of semantic assets that may affect the interpretation of ongoing tasks.
- "Manage agent discovery" capability of the "Agent management service" from Annex A. Used by the requesting agent to discover the target agent and retrieve its agent capability profile.
- "Manage subscriptions" capability of the "Management communication service" from ETSI GS ZSM 002 [1]. Used to manage subscriptions to semantic asset change notifications.

5.4.4 Requirements

The following requirements need to be fulfilled by the ZSM framework to enable semantic interoperability between autonomous agents:

Req-5.4.4-01: The ZSM framework shall support semantic interoperability between autonomous agents using different domain-specific terminology, information models or abstractions.

Req-5.4.4-02: The ZSM framework shall support semantic interoperability independently of the agent interaction protocol used for message exchange.

Req-5.4.4-03: The ZSM framework shall support the use of machine-readable semantic assets for agent interaction.

NOTE: Example of semantic assets are shared vocabularies and ontologies for defining concepts and relationships, semantic mappings between concepts or information models used by different management domains, and the other semantic assets described in clause 5.4.1.

Req-5.4.4-04: The ZSM framework shall support the inclusion of semantic metadata in agent messages and delegated tasks.

Req-5.4.4-05: Semantic metadata shall not modify or replace the primary content of the agent message.

Req-5.4.4-06: The ZSM framework may include support for a semantic repository for storing and exposing governed semantic assets.

Req-5.4.4-07: Where a semantic repository is deployed, the ZSM framework shall support retrieval of semantic assets required to interpret agent interactions.

6 Management domain support for Agents

6.1 Overview

As described in the preceding clauses, autonomous agents may be deployed as management functions within individual management domains and within the E2E service management domain. The specific functional capabilities of an agent depend on the management scope, managed entities, management services and operational objectives associated with the domain in which the agent operates.

The introduction of agents does not modify the separation of concerns between management domains defined in ETSI GS ZSM 002 [1]. Each management domain remains responsible for the management of its associated managed entities and controls the exposure of management services beyond its domain boundaries. Autonomous agents operate within these existing domain boundaries and consume and produce ZSM management services according to the policies and authorization rules applicable to the respective management domain.

Work in other standards and research organizations is increasingly introducing AI and autonomous-agent capabilities within technology-specific management domains. These activities do not necessarily define a common agent architecture, since the detailed implementation remains domain and technology specific. However, they increasingly exhibit common interaction patterns that are aligned with the agent management and communication mechanisms defined in the present document, including:

1) agent capability or skill advertisement;
2) agent discovery;
3) high-level goal or task delegation;
4) local autonomous reasoning and execution;
5) status and result exchange; and
6) cross-agent coordination.

The following clauses provide examples of such related industry work.

6.2 Agents in 3GPP Core domain and RAN domain

3GPP is studying the application of AI/ML and agentic AI to the management and operation of mobile networks.

In NG-RAN, AI/ML-related work addresses use cases including network energy saving, load balancing, mobility optimization and other network optimization functions.

In the 6G Core and management architecture, ongoing Rel-20 studies consider agentic AI for autonomous OAM, intent-driven operation, AI-agent registration and discovery, agent collaboration, semantic capability discovery and AI-agent-based network control [i.9].

These activities provide examples of domain-specific agent functional capabilities that may be exposed, discovered and coordinated using the common agent mechanisms defined in clause 5.

6.3 Agents in IP and Transport domain

The IRTF Network Management Research Group (NMRG) is investigating the use of agentic AI for autonomous IP and transport network management. Current work includes architectures for agentic intent-based networking, AI-agent coordination, application of the A2A protocol to multi-domain network management, Controller-Agent and Device-Agent interaction, and the integration of Network Digital Twins with agentic AI for intent-driven closed-loop operations [i.10], [i.11].

These approaches illustrate a transition from imperative device management towards the delegation of higher-level management goals and tasks to autonomous agents. Existing network management protocols, such as NETCONF, RESTCONF and gNMI, may continue to be used for execution of device-level operations. These activities are currently documented in research Internet-Drafts and therefore represent work in progress rather than established IETF standards.

6.4 Agents in Fixed access domain

ETSI ISG F5G is applying AI and autonomous-network concepts to fixed access and optical network management. ETSI GR F5G 032 [i.12] defines a Generative AI use case for F5G-Advanced network management in which LLM-based agents can interpret operator intents, generate management actions, and automate configuration and monitoring tasks across PON, OTN and IP/Ethernet fixed networks. The use case also considers realization through a multi-agent architecture.

F5G work additionally addresses distributed intelligence for FTTR and autonomous fixed-network operation. Proof-of-Concept activities have demonstrated domain-specialized AI agents collaborating on configuration, inventory, service provisioning, telemetry and energy-management tasks in IP/optical networks, as well as local intelligent functions performing autonomous FTTR monitoring and reconfiguration.

These activities provide examples of fixed-access and optical-network agent functional capabilities that may be integrated using the common agent mechanisms defined in clause 5.

6.5 Cross-domain and end-to-end agent scenarios

Ongoing work in ETSI ISG ZSM includes the definition of cross-domain and end-to-end agent scenarios that complement the architectural mechanisms specified in the present document.

Examples include:

- ETSI DGS/ZSM-023 (work in progress), Agent for Predictive and Cross-Domain Network Assurance;
- ETSI DGS/ZSM-025 (work in progress), AN Agent for Home Broadband Installation and Assurance;
- ETSI DGS/ZSM-026 (work in progress), Framework for Intent-Driven Operation Orchestration in Agent-Based E2E Service Management Domain; and
- ETSI DGS/ZSM-030 (work in progress), AN Agent for End-to-End Network Optimization.

These activities illustrate how autonomous agents may collaborate across management domains and within the E2E service management domain to support assurance, fulfilment, orchestration and optimization scenarios. They provide concrete examples of the cross-domain discovery, task delegation, coordination, semantic interoperability and closed-loop interaction mechanisms defined in clause 5.

Annex A (normative): Management Services

A.1 Additional Services for ZSM architecture

A.1.1 Overview

This clause describes the additional ZSM services required in the integration fabric to support agent integration within the ZSM framework:

- Agent management service:
  - Capability for agent registration/de-registration: Enables the lifecycle management of agent identities and capabilities within the agent registry.
  - Capability for agent discovery: Enables consumers to identify registered agents and their access endpoints based on semantic capability matching.
- Semantic interoperability service: management of semantic assets.

The following services are required in management domains and the E2E service management domain:

- Agent communication service: supports standardized agent message exchange and task lifecycle management between collaborating agents.
- The capabilities offered by these additional services are defined under "Service capabilities" in the following clauses. A capability can be Mandatory (M) which means it shall be offered if the service is offered, or Optional (O) which means it may be offered if the service is offered.

A.1.2 Agent management service

The Agent management service manages the registration, update and de-registration of autonomous agents in the agent registry that can be discovered using the agent discovery service. It allows authorized consumers to manage (create, read, update, delete, list) agent capability profiles in the registry.

Additionally, it enables the ZSM framework consumers to discover the available agents and their supported functional capabilities.

The Agent management service complements the management service registration and discovery mechanisms defined in ETSI GS ZSM 002 [1] by managing agent-specific information. While the management services produced by an agent and their associated service capabilities are registered and discovered using the mechanisms defined in ETSI GS ZSM 002 [1], the Agent management service manages the agent capability profile, including the agent's functional capabilities, lifecycle information and supported interaction mechanisms, and enables discovery of agents based on these agent-specific characteristics.

The service is further detailed in table A.1.2-1.

Table A.1.2-1: Service definition

Service name		Agent management service
Service capabilities		
	Manage agent registration (M)	Manage (create, read, update, delete, list) the registration of agents in the agent registry.
	Manage agent discovery (M)	Enable the identification and retrieval of registered agents within the agent registry, based on semantic understanding of the functional capabilities declared in the agent capability profile. This function may leverage structured metadata and ontological models to interpret the meaning and relationships of agent functional capabilities allowing discovery not just by name or type, but by the functional abilities, intents, and dependencies an agent exposes (see note).
	Manage agent lifecycle (M).	Manage the lifecycle of agents. The set of applicable lifecycle management operations depends on the agent type, but typically includes the instantiation, activation, de-activation and termination of the agent.
	Provide notifications of agent changes (M)	Notify subscribed consumers about changes in the agent registry, including lifecycle operations and any resulting updates to the affected agents.
NOTE: It supports semantic queries (e.g. "find agents able to perform anomaly detection in transport networks") and returns results that align with both explicit agent functional capability descriptions and inferred capabilities.		

A.1.3 Agent communication service

The Agent communication service enables the sending of standardized agent messages between ZSM authorized agents. This capability accepts objects containing message content, metadata and configuration. Depending on the interaction pattern, the service may return a task representation for asynchronous processing or a direct agent-message response for synchronous interactions.

The Agent communication service defines agent-specific communication capabilities, including agent messages and task lifecycle management. It may use the communication mechanisms provided by the Management communication service defined in ETSI GS ZSM 002 [1] for message delivery, including publish-subscribe communication.

The service is further detailed in table A.1.3-1.

Table A.1.3-1: Service definition

Service name		Agent communication service
Service capabilities		
	Provide agent message (M)	The primary means for agents to exchange intents and task details (see note).
	Manage task (M)	Manage (create, read, update, delete, list) tasks.
	Provide task status (M)	Notify subscribed consumers about changes in the status of tasks.
NOTE: Supports both peer-to-peer targeting via agent endpoints from the agent registry and publish-subscribe via topics, with semantic routing based on intent and functional capability matching.		

A.1.4 Semantic interoperability service

The semantic interoperability service is an optional management service provided by the integration fabric. It enables authorized consumers to store, retrieve, validate and resolve semantic assets used to ensure consistent interpretation of goals, intents, tasks, events, context items and expected outputs exchanged between agents operating in different management domains.

When deployed, the semantic interoperability service acts as the governed source of semantic assets for the ZSM multi-agent framework. It supports the semantically-enriched task delegation process defined in clause 5.4.3 and may be used by the integration fabric to perform semantic validation and mediation transparently during message routing.

The service is further detailed in table A.1.4-1.

Table A.1.4-1: Service definition

Service name		Semantic interoperability service
Service capabilities		
	Manage semantic assets (O)	Manage (create, read, update, delete, list) semantic assets in the semantic repository. Semantic assets include shared vocabularies, ontologies, taxonomies, semantic mappings, metric definitions, unit definitions and model references. Read and list operations shall be supported. Create, update and delete operations may be supported.
	Validate semantic reference (O)	Validate semantic references used by the agents. Returns a validation result including any identified inconsistencies or unknown references.
	Resolve semantic reference (O)	Resolve the meaning of a semantic reference, returning the applicable definition, mapping or equivalence for a given term. Supports semantic queries that may return direct definitions as well as inferred or mapped equivalences from stored ontologies, taxonomies and semantic mappings (see note).
	Provide semantic asset notifications (O)	Notify subscribed consumers about changes to semantic assets.
NOTE: The "Resolve semantic reference" capability supports semantic queries during agent discovery (clause 5.2.4) and interpretation of semantic metadata in agent messages (clause 5.4.2.2).		

Annex B (informative): Multi-agent ZSM scenarios

B.1 End-to-end zero-touch slice lifecycle using multiple agents

B.1.1 Use case description

This use case illustrates how autonomous agents discover one another and coordinate operations to manage the lifecycle of an end-to-end network slice using the **Agent management service** and **Agent communication service** defined in the present document.

The use case is based on the network-slicing management concepts defined by 3GPP. ETSI TS 128 530 [i.8] defines a NetworkSlice instance representing the service view of a network slice and NetworkSliceSubnet instances representing sets of network functions and associated resources supporting the network slice. A NetworkSlice instance may contain constituent network slice subnets belonging to different technology domains.

In the agent-based realization illustrated in this use case, an **E2E service management domain agent** coordinates the lifecycle of the end-to-end network slice. Domain agents operating in the RAN, Transport and Core Network management domains expose functional capabilities related to the lifecycle management of the corresponding network slice subnets.

The **Agent management service** enables the E2E service management domain agent to discover the agents capable of performing the required domain-specific slice lifecycle operations. The **Agent communication service** enables the E2E service management domain agent and the selected domain agents to exchange lifecycle tasks, task status and execution results.

B.1.2 Preconditions

The following preconditions apply:

- The E2E service management domain agent is registered and active.
- The domain agents participating in the slice lifecycle are registered and active.
- The agent capability profiles of the participating domain agents identify their applicable functional capabilities and interaction endpoints.

B.1.3 Interaction flow

The detailed steps for this use case are the following.

1) **Slice lifecycle request**

   A ZSM consumer requests the creation of an end-to-end network slice from the E2E service management domain agent. The request contains the information required by the E2E agent to determine the required slice constituents and lifecycle operation.

   The requested operation may correspond to one of the lifecycle operations defined by 3GPP for a slice instance, such as creation, activation, modification, de-activation or termination.

2) **Identification of required domain capabilities**

   The E2E service management domain agent determines the management domain capabilities required to fulfil the requested slice operation.

For example, the creation of a new end-to-end network slice may require creation or modification of RAN and Core Network slice subnet instances and the establishment or modification of the required Transport connectivity.

3) **Agent discovery**

The E2E service management domain agent consumes the "**Manage agent discovery**" capability of the **Agent management service** to identify domain agents exposing the required slice lifecycle functional capabilities.

The discovery process may identify RAN, Transport and Core Network management domain agents capable of managing network slice subnets and required connectivity.

The agent discovery mechanism defined in clause 5.2 returns agent capability profiles and applicable interaction endpoints that can subsequently be used for task delegation.

4) **Domain task delegation**

The E2E service management domain agent creates domain-specific tasks corresponding to the required lifecycle operation and delegates them to the selected domain agents using the **"Provide agent message"** capability of the **Agent communication service**.

For example, when commissioning a new network slice, the E2E agent may delegate:

- creation of the required RAN slice subnet instance to the RAN-domain agent;
- creation of the required Core Network slice subnet instance to the Core-domain agent; and
- establishment or modification of the required transport connectivity to the Transport-domain agent.

5) **Task acceptance**

Each domain agent evaluates the delegated task against its registered functional capabilities and current operational conditions:

- A domain agent that can perform the requested operation accepts the task and returns the corresponding task status to the E2E service management domain agent.
- If a domain agent cannot perform the requested operation, it may decline the task. The E2E agent may subsequently use the Agent management service to discover another suitable agent.

6) **Domain-specific lifecycle execution**

Each domain agent performs the accepted lifecycle task by consuming the appropriate management services within its management domain. The implementation of these domain-specific operations is outside the scope of the present document.

7) **Task status reporting**

During execution, the domain agents provide task-status updates to the E2E service management domain agent using the Agent communication service. The E2E agent uses these updates to maintain an end-to-end view of the lifecycle operation across the participating management domains.

8) **Domain task completion**

Once a domain lifecycle operation has completed, the corresponding domain agent reports the task outcome to the E2E service management domain agent. The result may identify the created or modified network slice subnet, its lifecycle state and any information required by the E2E agent to continue the overall slice lifecycle process.

9) **End-to-end coordination**

The E2E service management domain agent correlates the task results received from the participating domain agents. Where dependencies exist between lifecycle operations, the E2E agent may sequence the corresponding tasks. For example, activation of the end-to-end network slice may occur only after the required constituent network slice subnets have been successfully created and made available.

10) **End-to-end completion**

After all required domain tasks have completed, the E2E service management domain agent reports successful completion of the requested lifecycle operation to the ZSM consumer.

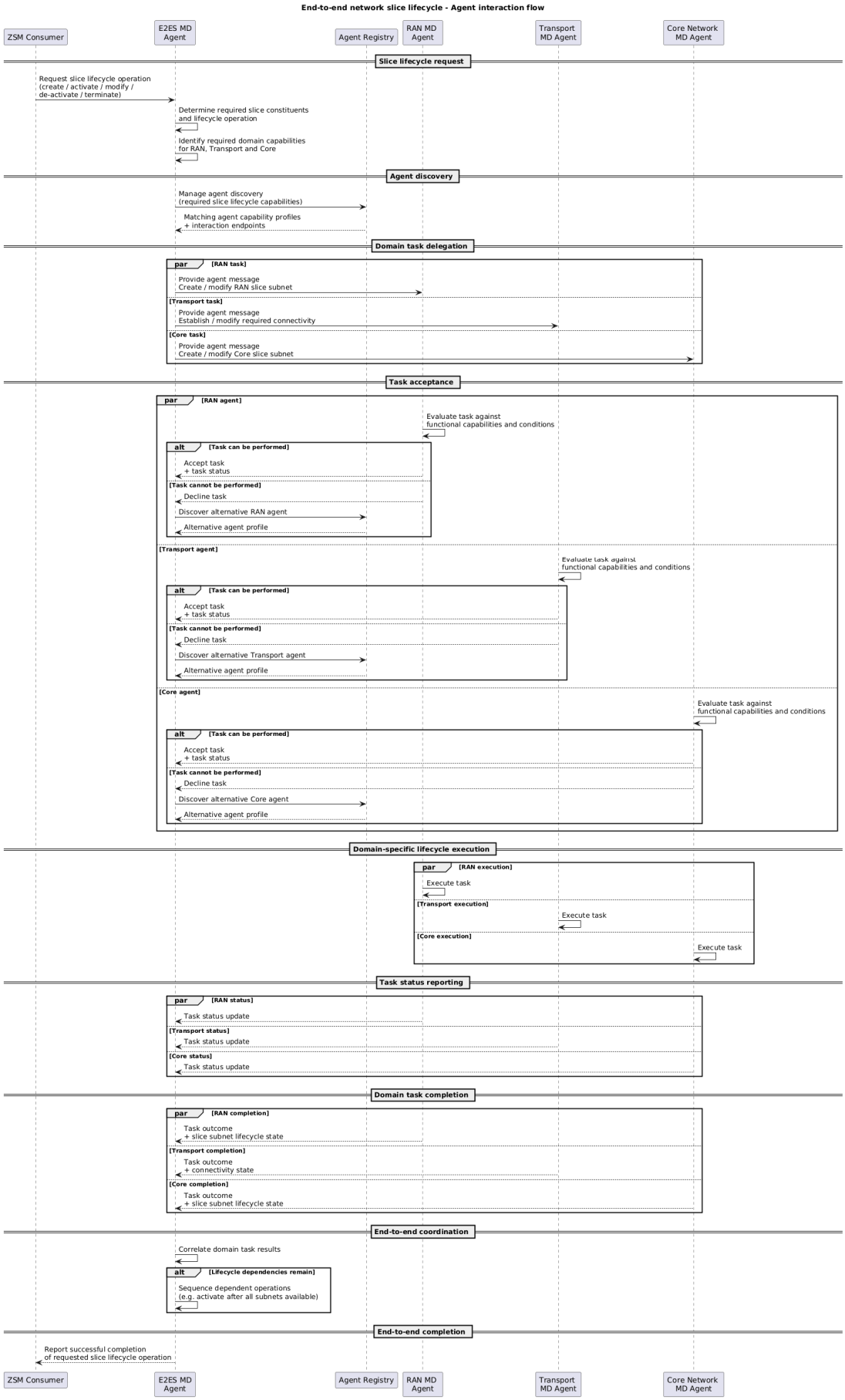


Figure B.1.3-1: Workflow for End-to-end zero-touch slice lifecycle using multiple agents

B.2 Cross-domain fault correlation and remediation

B.2.1 Use case description

This use case is defined in ZSM PoC17. It demonstrates how multiple agents collaborate to correlate and remediate faults detected across multiple management domains while minimizing the impact on service experience.

B.2.2 Preconditions

- All agents have registered their capabilities in the Agent Registry, and can be discovered based on their registered functional capabilities.

B.2.3 Interaction flow

The detailed steps for this use case are the following:

1) The ZSM consumer delivers a service assurance intent ("service assurance for gold-tier customer") to E2E service management domain agent, with the intent expectation on radio network performance (e.g. RAN UE throughput performance in the specific area).

2) E2E service management domain agent analyses intent, decomposes the intent into domain-specific sub-intents, and forms a collaboration team with related management domain agents (RAN management domain agent, transport management domain agent) by discovering agents from the Agent Registry.

3) The E2E service management domain agent starts monitoring service quality and assigns tasks to the participating management domain agents. These tasks can include sub-intents, subscription information, etc.

4) RAN management domain agent detects an anomaly ("gNB-BJ-001 base station outage leads to widespread packet loss in regional ping tests."). After determining that the root cause is not within the RAN management domain, it immediately performs automated compensation actions and reports them to the E2E service management domain agent.

5) Concurrently, the transport management domain agent detects an anomaly ("BER alarm on SPE cross-connect optical cable; primary SPE-to-NPE physical link down"). After analysis, it diagnoses that the root cause lies within its domain and executes an emergency repair (e.g. switch the UPE to the standby SPE) immediately, while also generating a recommended on-site repair solution (e.g. repair the SPE primary optical fiber) and reporting the full details to the E2E service management domain agent.

6) The E2E service management domain agent receives several fault reports from RAN management domain and transport management domain, and detects the degradation of service quality. It correlates these issues by cross-domain demarcation analysis, and identifies that the root cause is in transport management domain, then determines the optimal repair solution.

7) The E2E service management domain agent formulates the repair tasks, and triggers a Field Maintenance Engineer (FME) dispatch for on-site remediation.

8) Upon detecting stable service quality and receiving fault resolution notification from the E2E service management domain agent, the RAN agent rolls back its compensation actions.

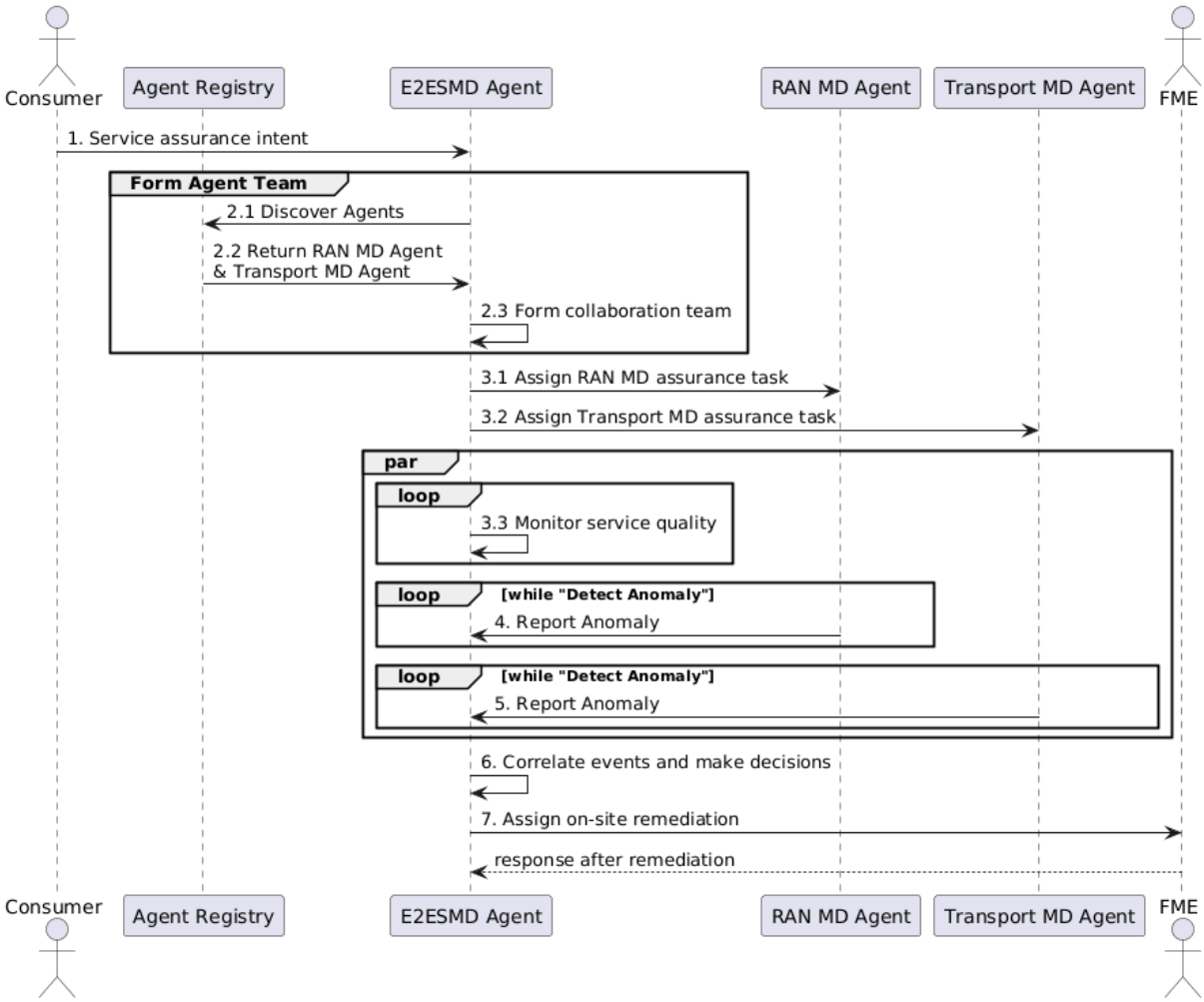


Figure B.2.3-1: Workflow for Cross-domain fault management

B.3 Closed-loop energy optimization

B.3.1 Use case description

This use case illustrates how an E2E service management domain agent and a RAN management-domain agent collaborate to fulfil a network energy-saving intent while preserving applicable service-performance constraints.

The E2E service management domain agent operates at a service-oriented level and receives an energy-efficiency objective expressed as an intent. The RAN management-domain agent operates within the RAN management domain and exposes functional capabilities for evaluating and executing energy-saving actions on the managed RAN resources.

Because the two agents operate at different management abstraction levels and may use different terminology and information models, a shared Autonomous Networks (AN) ontology is used to establish a common semantic understanding of the intent and of the information exchanged during task delegation and negotiation.

The shared ontology does not require the agents to use the same internal implementation, AI model or domain information model. Instead, it provides the semantic reference used for exchanged information. Domain-specific information can therefore remain internal to the RAN management domain while the information required for cross-domain collaboration is represented using commonly understood concepts.

The use case demonstrates the semantic-interoperability mechanisms described in clause 5.4, including the use of shared ontologies, semantic metadata and clarification exchanges. Clause 5.4 already allows semantic references, source and target domains, metric definitions, constraint semantics and expected-output semantics to accompany agent messages.

B.3.2 Preconditions

The following preconditions apply:

- The E2E service management domain agent and the RAN management-domain agent are registered and can be discovered using the mechanisms specified in clause 5.2.
- The RAN management-domain agent advertises an agent functional capability related to RAN energy optimization or energy saving.
- Both agents have access to, or can resolve references to, a common AN ontology containing the semantic concepts required for the interaction.
- The common ontology contains the generic intent concepts used between the agents and references or mappings to the RAN energy-saving domain ontology.

B.3.3 Interaction flow

The detailed steps for this use case are the following:

1) **Energy-saving intent creation:**
A ZSM consumer provides the E2E service management domain agent with an energy-saving intent. The intent may initially be expressed using natural language, structured information, or a combination of both. For example:

"Reduce energy consumption in Region 1 while maintaining at least 20 Mbps user throughput and without degrading user experience."

2) **Semantic interpretation of the intent:**
The E2E service management domain agent interprets the intent using the AN ontology and identifies the corresponding semantic concepts and values. In the example above, these include an Energy-Saving Region, a Rate Guarantee Target and an Experience Lossless Energy-Saving Required condition.

3) **Agent discovery:**
The E2E service management domain agent discovers a RAN management-domain agent exposing the required energy-saving functional capability using the agent discovery mechanisms defined in clause 5.2.

4) **Semantically-enriched task delegation:**
The E2E service management domain agent delegates the energy-saving task to the RAN management-domain agent using the mechanisms specified in clause 5.3. The message includes the original intent or task together with semantic metadata identifying the applicable AN ontology and the relevant energy-saving concepts.

The semantic information may describe, for example:

- energy-saving region = Region1
- rate guarantee target = 20 Mbps
- gain target = 5 %
- coverage type = All
- energy-saving mode = LTE, NR, LNR or All
- energy-saving aggressiveness = Conservative or Aggressive
- lossless user experience required = Yes
- energy-saving time = 22:00 - 06:00

5) **Semantic interpretation by the RAN agent:**
The RAN management-domain agent interprets the task according to the referenced ontology and maps the received semantic concepts to the corresponding domain-specific capabilities, managed entities, policies and parameters.

6) **Target clarification, where required:**
If one or more intent concepts are ambiguous or incomplete, the RAN management-domain agent requests clarification from the E2E service management domain agent. For example, if the intent requests *"aggressive energy saving without service degradation"* but does not provide a minimum throughput requirement, the RAN agent may request the missing Rate Guarantee Target.

The clarification exchange uses the same ontology concepts so that the requested information and the response have an agreed meaning.

7) **Feasibility assessment:**
Once the intent is semantically understood, the RAN management-domain agent evaluates whether the requested energy-saving objective can be fulfilled under the specified constraints.

The RAN agent may use information from its management domain, such as current traffic load, coverage conditions, resource availability and service-performance information, to perform the feasibility assessment.

8) **Feasibility negotiation:**
If the original intent cannot be fulfilled exactly, the RAN management-domain agent may return a counter-proposal expressed using the same ontology.

For example, if an aggressive energy-saving strategy cannot simultaneously satisfy a lossless-experience requirement and a 20 Mbps minimum throughput requirement, the RAN agent may propose:

- reducing the energy-saving aggressiveness from Aggressive to Conservative;
- reducing the requested energy-saving gain;
- modifying the active time window; or
- restricting the set of cells participating in the energy-saving operation.

9) **Intent commitment:**
When the E2E service management domain agent and the RAN management-domain agent reach agreement on the objective and constraints, the RAN agent accepts the delegated task and commits to the resulting energy-saving intent.

The agreed intent is represented using the shared semantic concepts so that both agents retain the same interpretation of the objective, constraints and expected outcome.

10) **Closed-loop execution within the RAN management domain:**
The RAN management-domain agent autonomously determines and executes the management actions required to satisfy the agreed intent. These implementation-specific actions remain within the RAN management domain and may involve one or more closed loops, management services or local optimization functions.

11) **Status and result reporting:**
During execution, the RAN management-domain agent provides task-status information to the E2E service management domain agent. After execution, the RAN agent reports the achieved energy-saving result and the corresponding service-performance information using semantic concepts and metric definitions that can be consistently interpreted by the E2E service management domain agent.

12) **E2E evaluation and adaptation:**
The E2E service management domain agent evaluates the reported outcome against the original E2E objective. If the expected energy-saving or service-quality objectives are not satisfied, it may modify the intent, initiate another negotiation cycle or coordinate additional management-domain agents.

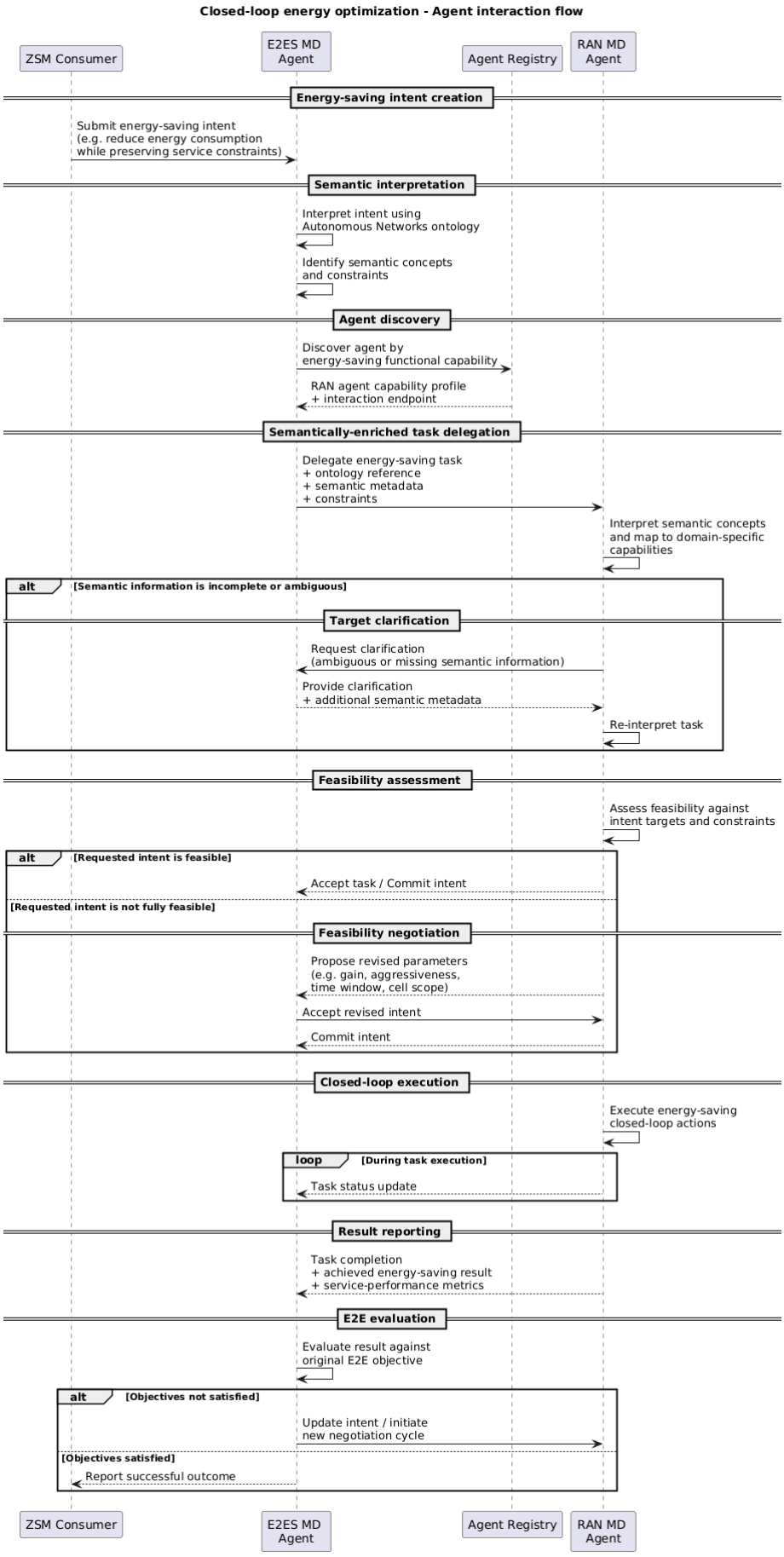


Figure B.3.3-1: Workflow for Closed-loop energy optimization

B.3.4 Role of the AN ontology

The AN ontology provides the shared semantic reference between the E2E service management domain agent and the RAN management-domain agent. It enables an intent expressed at the E2E service management level to be related to the domain-specific concepts understood by the RAN agent without requiring the E2E agent to understand the internal implementation of the RAN management domain.

The ontology may be structured in layers. A common ontology layer defines reusable concepts such as Intent, Expectation, Target and Condition, while domain-specific ontology extensions specialize these concepts for particular management capabilities.

In the energy-saving example, the RAN-specific ontology defines concepts such as Energy-Saving Region, Rate Guarantee Target, Gain Target, Energy-Saving Aggressiveness and Energy-Saving Time. This layered approach allows different management domains to extend the common semantic model with their own domain-specific concepts while retaining semantic relationships to the shared AN ontology.

The cross-domain interaction therefore exchanges the semantic information necessary for shared understanding, while domain-specific knowledge and detailed operational information remain within the corresponding management domain.

This supports consistent interpretation by agents that may use different implementations or AI models. The shared ontology enables alignment of the semantics used in requester and executor messages, thereby supporting consistent understanding, reasoning and execution.

Annex C (informative): Examples of semantic data model

C.1 Description

The present document introduces messages and tasks as units of information exchanged between agents, conveying goals, intents, tasks, events, context, constraints, and outcomes. Such exchanged information may refer to a Closed-Loop (CL) which may exist in any of the management domains of the ZSM architecture. Structured prompts are one example of realizing semantic metadata, described in clause 5.4.2.2, which play a crucial role in guiding autonomous agents to perform tasks effectively. This clause provides typical examples of effective prompts that agents use to express a message or task conveyed between CLs or CL stages.

Due to the high readability, structured organization, and ease of format conversion of Markdown, this clause utilizes Markdown format to introduce the metamodel sample of structural prompts. The specific format is as follows: using "##" as the identifier for metamodel properties, followed by entering the name of the metamodel property.

C.2 Data model for task

Since an autonomous agent is defined as an intent-driven, autonomous AI entity, the intent carried within a task is a major case in agent-to-agent communication. Leveraging natural language can improve the flexibility in processing abstract tasks, but introduces ambiguity in agent-to-agent communications. Therefore, a data model for structured prompts can improve robustness. Table C.2-1 lists attributes for expressing tasks by the structured prompt, which can offer accuracy and stability advantages for deterministic task execution and precise information specification.

Table C.2-1: Attributes of data model for intent-related structured prompt

Task Description	Used in messages sent from the sender agent (the task requester) to the receiver agent (the task executor) to describe the fundamental requirements of the task.
Task Type	An optional attribute used to define the type of the task. Typical task type include optimization, analysis, complaint handling, etc.
Task Object	An optional attribute used to express the object's identifier and related information, such as "all network slices in city ABC", "work order XXX".
Task Context	An optional attribute used to define the environmental context related to the task, such as the time and location of a fault occurrence in a fault management scenario, or additional contextual information like neighbouring management domains or related agent details.
Constraints	An optional attribute used to define the priority of the task to help resolve conflicts among task objectives.
Expected Output	Defines the format of the anticipated result. It is recommended to include whether the task was successful, records of task operations, and suggested actions for manual intervention.

An example of a Fault Repair Task Prompt Template is shown below:

```
## Task Description
Please resolve the fault based on the following fault information.

## Task Type
Fault Repair

## Task Object
Includes elements such as a list of fault serial numbers.

## Task Context
Includes elements such as the time of occurrence, location, and fault context.

## Constraints
Includes elements such as priority and the content of the task result output.
```

Expected Output
The fault resolution result should include the following information:
1. Resolution Result: Success or Failure
2. Fault Resolution Details
3. Reason for Resolution Failure (if applicable)
4. Domain-Specific Information

C.3 Compatibility with Intent data model

To ensure compatibility, the intent model (e.g. as defined in ETSI TS 128 312 [i.3]) can be carried in the data block of a task.

Taking the ETSI TS 128 312 [i.3] intent model as an example, the request body is shown below:


```
{
 "messageId": "9229e770-767c-417b-a0b0-f0741243c589",
 "message": {
  "role": "user",
  "parts": [
   {
    "data": {
     "Intent": {
      "Id": "Intent_5",
      "userLabel": "RAN_Energy_Saving",
      "intentExpectation": [
       {
        "expectationId": "1",
        "expectationVerb": "Ensure",
        "expectationObjects": [
         {
          "objectInstance": "SubNetwork_1",
          "objectContexts": [
           {
            "contextAttribute": "CoverageAreaPolygon",
            "contextCondition": "IS_ALL_OF",
            "contextValueRange": [
             {
              "convexGeoPolygon": [
               { "latitude": "31.2696", "longitude": "121.6322" },
               { "latitude": "31.2668", "longitude": "121.6323" },
               { "latitude": "31.2669", "longitude": "121.6412" },
               { "latitude": "31.2696", "longitude": "121.6410" }
              ]
             }
            ]
           }
          ]
         }
        ],
        "expectationTargets": [
         {
          "targetName": "RANEnergyConsumption",
          "targetCondition": "IS_LESS_THAN",
          "targetValueRange": "1000"
         },
         {
          "targetName": "AveULRANUEThpt",
          "targetCondition": "IS_GREATER_THAN",
          "targetValueRange": "100"
```

```
            },
           ],
           "expectationContexts": [
            {
             "contextAttribute": "TargetAssuranceTime",
             "contextCondition": "IS_EQUAL_TO",
             "contextValueRange": [
              { "startTime": "2023-10-27-22-00-00" },
              { "endTime": "2023-10-28-06-00-00" }
             ]
            }
           ]
          }
         ]
        }
       }
      }
    ]
  }
}
```

C.4 Data model for event

In a self-healing scenario, the fault event service provides notification capability to report events. A structured prompt-based network event subscription and reporting mechanism is needed, to ensure real-time perception of network events and provide consistent task descriptions across different layers and domains. Various standards organizations have defined notification mechanisms, for instance:

- IETF RFC 8639 [i.5] defines a general mechanism for subscribing to event notifications defined in YANG, supporting both dynamic and configured subscriptions.
- ETSI TS 128 312 [i.3] defines intent reports to notify intent fulfilment and status changes.
- A2A [i.6] employs push notifications for task updates.

This clause describes a data model based on natural language patterns, and is independent of the transport mechanism. Table C.4-1 lists attributes for expressing event notification with a structured prompt.

Table C.4-1: Attributes of data model for event-notification in structured prompt

Subscription Description	Used by the producer agent (the publisher of a network event) to send messages to the consumer agent (the subscriber of a network event), describing the basic requirements for event subscription and notification.
Notification Subject	An optional attribute used to define the types of events to be subscribed and notification, such as incident, energy efficiency optimization events, etc.
Notification Conditions	An optional attribute used to define the conditions for triggering the notification.
Event Data Reporting Format	An optional attribute used to define the expected format for event reporting to facilitate linking related tasks.
Expected Output	Defines the format of the anticipated result. It indicates whether the subscription and notification were successful.

An example of a Network Incident subscription and notification Prompt Template is shown below:

```
## Subscription Description
Please complete the subscription and reporting of autonomous data based on the following information: event topic, subscription criteria, expected output format, etc.

## Notification Subject
incident
```

Notification conditions
Report when the fault level reaches：Primary | Intermediate | critical

Event Data Reporting Format
Incident reporting should include the following information:
- Basic incident information
Includes message type, fault ID, fault name, fault time, fault type, fault location, etc.
- Incident analysis results
Results of incident analysis and diagnosis, such as root cause of the fault and diagnosis details.
- Event Impact Level
Includes information such as fault level, impact of the fault, etc.
- Event Handling Recommendations
- Fault Resolution Suggestions

Expected Output
1. Subscription status, success or failure
2. Reason for subscription failure (optional)

C.5 Data model for task negotiation

Task negotiation encompasses disambiguation of tasks (target negotiation), resolution of resource conflicts (feasibility negotiation), and inquiries for necessary information during task execution (information negotiation). For example, feasibility negotiation refers to a discussion about the feasibility of the task objectives, aiming to avoid issues such as resource unavailability - e.g. when the required collaborating agent is currently occupied with other tasks and thus unavailable. Table C.5-1 lists attributes used for ambiguity resolution of task objectives.

Table C.5-1: Attributes of data model for task-negotiation in structured prompt

Objective Confirmation	Used by the task executor agent to send a request to the task requester agent for objective negotiation, addressing ambiguity in the task goal.
Original Objective	Sent from the task requester to the task executor agent, with values described in natural language or structured data.
Objective Recognition	A summarized task description identified by the task executor agent.
Follow-Up Information	Optional information requested by the task executor to resolve ambiguity or provide missing information required to evaluate or execute the task.
Expected Output	When the original goal is ambiguous or lacks necessary details, the task executor agent sends a message to the task requester agent to supplement essential information.

An example of a Network Target Negotiation Prompt Template is shown below:

Intent Confirmation
Please confirm whether the intent recognition result is accurate. If not, provide additional information to clarify the intent.

Original Intent
The original input content (such as questions, commands, requests, etc.)

Intent Recognition Result
A brief summary of the intent

Follow-up information on Intent
The original intent is ambiguous; please provide precise follow-up information.

Expected Output
Please respond with the follow-up information regarding the intent.

C.6 Example of using data model in information exchange

To illustrate the use of the data model, the following example shows a request body using the A2A Protocol [i.6] together with the A2A-T telecom extension [i.13].


```
https://projects.tmforum.org/a2aproject/telecommunication/extensions/TASK-T/NL/v1
Body:
{
   "message": {
      "role": "agent",
      "parts": [
         {
            "text": "Fix the base station service outage in City X on January 16."
         }
      ],
      "metadata": {"https://projects.tmforum.org/a2aproject/telecommunication/extensions/TASK-T/NL/v1": "## Task Description\nPlease repair the fault based on the following fault information.\n## Task Type\nFault Repair\n## Task Object\nFault serial number list: 150285, 488703\n## Task Context\nLocation: X City\nTime: 2024-01-16 16:56:32\nFault Context: 4G base station outage fault.\n## Constraints\n\n## Expected Output\nThe fault resolution result should include the following information:\n1. Resolution Result: Success or Failure\n2. Fault Resolution Details\n3. Reason for Resolution Failure\n4. Domain-Specific Information\n"},
      "messageId": "9229e770-767c-417b-a0b0-f0741243c589"
   }
}
```

Annex D (informative): Change history

Date	Version	Information about changes
March 2026	V0.0.1	Initial skeleton
March 2026	V0.0.2	Incorporated contributions: • ZSM(26)000029r1_ZSM022_5_2_Agent_Onboarding • ZSM(26)000030r1_ZSM022_5_2_3_Process_Agent_Registration • ZSM(26)000031r1_ZSM022_5_2_4_Process_Agent_Discovery • ZSM(26)000033r2_ZSM022_5_1_Architectural_Overview • ZSM(26)000035r1_ZSM022_5_3_4_Process_Task_status_update__escalation_and_comp
May 2026	V0.0.3	Incorporated contributions: • ZSM(26)000032r2_ZSM022_5_3_Agent_Communication • ZSM(26)000034r2_ZSM022_5_3_3_Process_Tasks_delegation_and_negotiation • ZSM(26)000058_ZSM022_Updated_architectural_diagram
August 2026	V0.0.4	Incorporated contributions: • ZSM(26)000082r2_ZSM022-Clause_4_design_principles • ZSM(26)000095_ZSM022_5.4.1_Semantic_Interoperability_Overview • ZSM(26)000096_ZSM022_5.4.2_Semantic Components_Ontology • ZSM(26)000097_ZSM022_5.4.2_Semantic Components_Metadata • ZSM(26)000098_ZSM022_5.4.2_Semantic Components_repo • ZSM(26)000099_ZSM022_5.4.3 Process Semantically-enriched task delegation • ZSM(26)000100_ZSM022_Semantic interoperability service • ZSM(26)000101_ZSM022_Clause_5_communication_overview_ • ZSM(26)000102_ZSM022_Annex_C_Semantic_data_model • ZSM(26)000103_ZSM022-Clause_4_Agent_concept • ZSM(26)000105_ZSM022_Clause_4_multi-Agent_Concept_
September 2026	V0.0.5	Incorporated contributions: • ZSM(26)000039r1_ZSM022-B_2_Cross-domain_fault_correlation_and_remediation • ZSM(26)000213r2_ZSM022_Agent_registration_vs_ZSM_service_registration • ZSM(26)000215_ZSM022_Clause41_requirements • ZSM(26)000216_ZSM022_Clause42_requirements • ZSM(26)000217r1_ZSM022_Clause43_requirements • ZSM(26)000218r1_ZSM022_Clause52_requirements • ZSM(26)000219r1_ZSM022_Clause53_requirements • ZSM(26)000220_ZSM022_Clause54_requirements • ZSM(26)000235_ZSM022_B.3 Closed-loop energy optimization • ZSM(26)000240_ZSM022_Clause6 • ZSM(26)000247_B.1 End-to-end zero-touch slice lifecycle using multiple agents
September 2026	V0.0.6	Incorporated contributions: • ZSM(26)000238r1_ZSM022_Chapter_5_2_2_3_and_5_2_2_4_ • ZSM(26)000267_ZSM022_Architecture diagram update

History

Version	Date	Status
V1.1.1	October 2026	Publication